

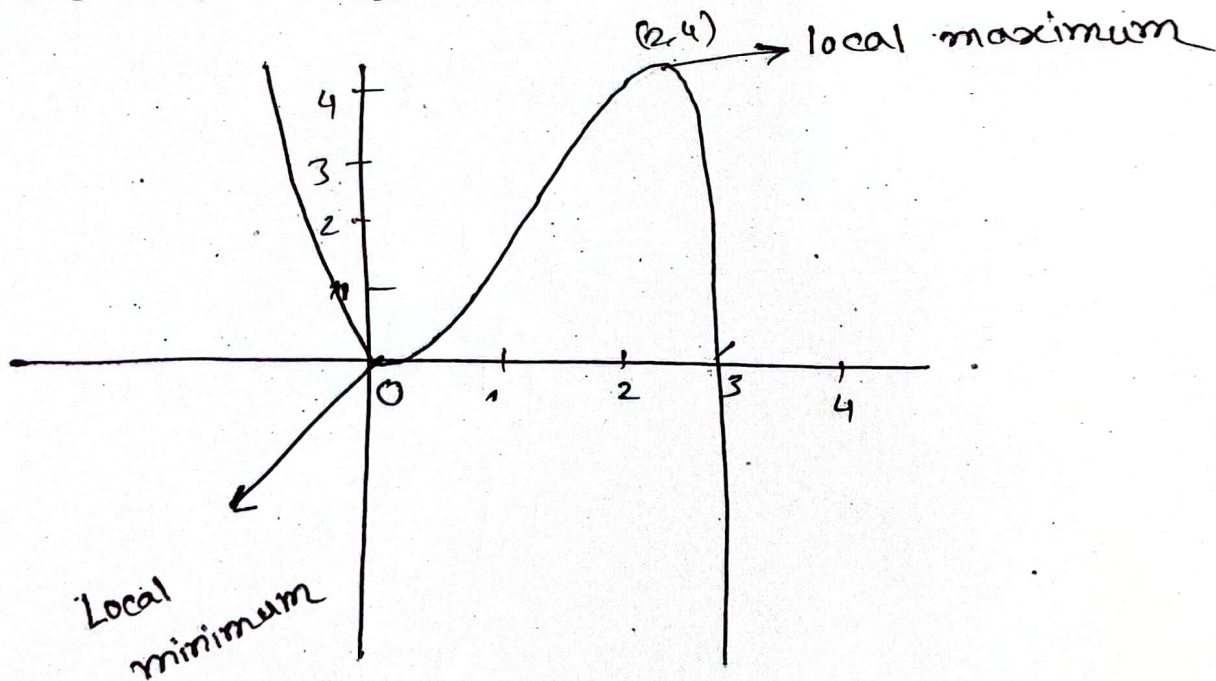
Differentiation

* What is Critical point of a graph? Give me an example with graph of a function.

$\Rightarrow$ All Sharp turning points are critical point.

Local minimum and local maximum.

$$f(x) = -x^3 + 3x^2$$



Condition for Maxima and Minima

Condition:

① The function $f(x)$ will be maximum

at $x = a$ if $f'(a) = 0$ and $f''(a) < 0$

② The function $f(x)$ will be minimum at

$x = a$ if $f'(a) = 0$ and $f''(a) > 0$.

③ The function $f(x)$ has neither maximum

nor minimum at $x = a$ if $f'(a) \neq 0$.

TV 00 1

1/ State and prove Euler's theorem on Homogeneous function.

Homogeneous function:- If the sum of the power of the each term of the function are equal is called Homogeneous function.

$$f(x, y) = x^n y^0 + x^{n-1} y^1 + \dots + y^n x^0$$

$$\text{Ex: } f(x, y) = a_0 x^n + a_1 x^{n-1} y + a_2 x^{n-2} y^2 + \dots + a_n y^n$$

$$= x^n [a_0 + a_1 \frac{y}{x} + a_2 (\frac{y}{x})^2 + \dots + a_n (\frac{y}{x})^n]$$

State of Euler's theorem

Statement:- If u be a homogeneous function

of degree n in x, y , we can write

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = n u$$

i) If $u = \sqrt{x^2 + y^2 + z^2}$ then prove that $u_{xx} + u_{yy} + u_{zz} = \frac{2}{u}$

Answer:

Given, $u = \sqrt{x^2 + y^2 + z^2}$ -----(i)

$$\therefore \frac{\delta u}{\delta x} = \frac{\delta}{\delta x} \sqrt{(x^2 + y^2 + z^2)}$$

$$\begin{aligned} &= \frac{\delta}{\delta x} (x^2 + y^2 + z^2)^{\frac{1}{2}} \\ &= \frac{1}{2} (x^2 + y^2 + z^2)^{\frac{1}{2}-1} \frac{\delta}{\delta x} (x^2 + y^2 + z^2) \quad \left[\because \frac{\delta}{\delta x} (x^n) = nx^{n-1} \right] \\ &= \frac{1}{2} (x^2 + y^2 + z^2)^{-\frac{1}{2}} \left\{ \frac{\delta}{\delta x} (x^2) + \frac{\delta}{\delta x} (y^2) + \frac{\delta}{\delta x} (z^2) \right\} \\ &= \frac{1}{2} (x^2 + y^2 + z^2)^{-\frac{1}{2}} (2x + 0 + 0) \\ &= \frac{1}{2(x^2 + y^2 + z^2)^{\frac{1}{2}}} \times 2x \end{aligned}$$

$$\therefore u_x = \frac{x}{\sqrt{(x^2 + y^2 + z^2)}}$$

Again differentiating

$$\begin{aligned} \frac{\delta}{\delta x} (u_x) &= \frac{\delta}{\delta x} \left(\frac{x}{\sqrt{(x^2 + y^2 + z^2)}} \right) \\ u_{xx} &= \frac{\sqrt{x^2 + y^2 + z^2} \frac{\delta}{\delta x} (x) - x \frac{\delta}{\delta x} \sqrt{(x^2 + y^2 + z^2)}}{(\sqrt{(x^2 + y^2 + z^2)})^2} \quad \left[\because \frac{\delta}{\delta x} \left(\frac{u}{v} \right) = \frac{v \frac{\delta}{\delta x} u - u \frac{\delta}{\delta x} v}{v^2} \right] \\ &= \frac{\sqrt{x^2 + y^2 + z^2} \times 1 - x \frac{\delta}{\delta x} \{(x^2 + y^2 + z^2)^{1/2}\}}{x^2 + y^2 + z^2} \\ &= \frac{\sqrt{x^2 + y^2 + z^2} - x \times \frac{1}{2} (x^2 + y^2 + z^2)^{\frac{1}{2}-1} \frac{\delta}{\delta x} (x^2 + y^2 + z^2)}{x^2 + y^2 + z^2} \\ &= \frac{\sqrt{x^2 + y^2 + z^2} - x \times \frac{1}{2} (x^2 + y^2 + z^2)^{-\frac{1}{2}} \left\{ \frac{\delta}{\delta x} (x^2) + \frac{\delta}{\delta x} (y^2) + \frac{\delta}{\delta x} (z^2) \right\}}{x^2 + y^2 + z^2} \\ &= \frac{\sqrt{x^2 + y^2 + z^2} - \frac{x}{2(x^2 + y^2 + z^2)^{1/2}} \{2x + 0 + 0\}}{x^2 + y^2 + z^2} \\ &= \frac{\sqrt{x^2 + y^2 + z^2} - \frac{x}{2(x^2 + y^2 + z^2)^{1/2}} \times 2x}{x^2 + y^2 + z^2} \\ &= \frac{\sqrt{x^2 + y^2 + z^2} - \frac{x^2}{(x^2 + y^2 + z^2)^{1/2}}}{x^2 + y^2 + z^2} \end{aligned}$$

$$\begin{aligned}
&= \frac{\sqrt{x^2 + y^2 + z^2} - \frac{x^2}{\sqrt{x^2 + y^2 + z^2}}}{x^2 + y^2 + z^2} \\
&= \frac{\frac{(\sqrt{x^2 + y^2 + z^2})^2 - x^2}{\sqrt{x^2 + y^2 + z^2}}}{x^2 + y^2 + z^2} \\
&= \frac{\frac{x^2 + y^2 + z^2 - x^2}{\sqrt{x^2 + y^2 + z^2}}}{x^2 + y^2 + z^2} \\
&= \frac{\frac{y^2 + z^2}{\sqrt{x^2 + y^2 + z^2}}}{x^2 + y^2 + z^2} \\
&= \frac{y^2 + z^2}{\sqrt{x^2 + y^2 + z^2}} \times \frac{1}{x^2 + y^2 + z^2} \\
u_{xx} &= \frac{y^2 + z^2}{(x^2 + y^2 + z^2)^{\frac{1}{2}}} \times \frac{1}{x^2 + y^2 + z^2} \\
&= \frac{y^2 + z^2}{(x^2 + y^2 + z^2)^{\frac{1}{2}+1}} \\
u_{xx} &= \frac{y^2 + z^2}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \text{-----(ii)}
\end{aligned}$$

Similarly

$$\begin{aligned}
u_{yy} &= \frac{x^2 + z^2}{(x^2 + y^2 + z^2)^{\frac{1}{2}}} \times \frac{1}{x^2 + y^2 + z^2} \\
u_{yy} &= \frac{x^2 + z^2}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \text{-----(iii)}
\end{aligned}$$

Similarly

$$\begin{aligned}
u_{zz} &= \frac{x^2 + y^2}{(x^2 + y^2 + z^2)^{\frac{1}{2}}} \times \frac{1}{x^2 + y^2 + z^2} \\
u_{zz} &= \frac{x^2 + y^2}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \text{-----(iv)}
\end{aligned}$$

Adding equation (ii), (iii) & (iv)

$$u_{xx} + u_{yy} + u_{zz} = \frac{y^2 + z^2}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} + \frac{x^2 + z^2}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} + \frac{x^2 + y^2}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}$$

$$u_{xx} + u_{yy} + u_{zz} = \frac{y^2 + z^2 + x^2 + z^2 + x^2 + y^2}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}$$

$$u_{xx} + u_{yy} + u_{zz} = \frac{2(x^2 + y^2 + z^2)}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}$$

$$u_{xx} + u_{yy} + u_{zz} = \frac{2(x^2 + y^2 + z^2)}{(x^2 + y^2 + z^2)^{\frac{1}{2}+1}}$$

$$u_{xx} + u_{yy} + u_{zz} = \frac{2(x^2 + y^2 + z^2)}{(x^2 + y^2 + z^2)^{\frac{1}{2}} \cdot (x^2 + y^2 + z^2)}$$

$$u_{xx} + u_{yy} + u_{zz} = \frac{2}{(x^2 + y^2 + z^2)^{\frac{1}{2}}}$$

$$u_{xx} + u_{yy} + u_{zz} = \frac{2}{\sqrt{(x^2 + y^2 + z^2)}}$$

$$u_{xx} + u_{yy} + u_{zz} = \frac{2}{u} \quad \text{Proved}$$

ii) If $u(x, y) = \frac{x^2 + y^2}{\sqrt{x + y}}$, Prove that

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{3}{2} u.$$

Solution:

$$u(x, y) = \frac{x^2 + y^2}{\sqrt{x + y}}$$

$$u(\lambda x, \lambda y) = \frac{\lambda^2 x^2 + \lambda^2 y^2}{\sqrt{\lambda x + \lambda y}}$$

$$= \frac{\lambda^2 (x^2 + y^2)}{\lambda^{1/2} \sqrt{x + y}}$$

$$= \frac{\lambda^{1/2} (x^2 + y^2)}{\sqrt{x + y}}$$

u is a homogeneous function of degree $\frac{3}{2}$

By Euler's Theorem,

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{3}{2} u$$

iii) Prove that $f(x, y) = x^3 - 2x^2y + 3xy^2 + y^3$ is homogeneous; what is the degree? Verify Euler's Theorem for f .

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$$

Solution:

$$f(x, y) = x^3 - 2x^2y + 3xy^2 + y^3$$

$$\begin{aligned} f(\lambda x, \lambda y) &= \lambda^3 x^3 - 2\lambda^2 x^2 \lambda y + 3\lambda x \lambda^2 y^2 + \lambda^3 y^3 \\ &= \lambda^3 (x^3 - 2x^2y + 3xy^2 + y^3) \end{aligned}$$

f is a homogeneous function of degree 3

By Euler's Theorem,

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = 3f$$

Verification:

$$f(x, y) = x^3 - 2x^2y + 3xy^2 + y^3$$

$$\frac{\partial f}{\partial x} = 3x^2 - 4xy + 3y^2$$

$$x \frac{\partial f}{\partial x} = 3x^3 - 4x^2y + 3xy^2$$

Verification:

$$f(x, y) = x^3 - 2x^2y + 3xy^2 + y^3$$

$$\frac{\partial f}{\partial x} = 3x^2 - 4xy + 3y^2$$

$$x \frac{\partial f}{\partial x} = 3x^3 - 4x^2y + 3xy^2$$

$$\frac{\partial f}{\partial y} = -2x^2 + 6xy + 3y^2$$

$$y \frac{\partial f}{\partial y} = -2x^2y + 6xy^2 + 3y^3$$

$$\begin{aligned} x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} &= 3x^3 - 4x^2y + 3xy^2 - 2x^2y + 6xy^2 + 3y^3 \\ &= 3x^3 - 6x^2y + 9xy^2 + 3y^3 \\ &= 3(x^3 - 2x^2y + 3xy^2 + y^3) \end{aligned}$$

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = 3f$$

We verified the Euler's Theorem.

iv)

Given,

$$u = \frac{1}{r} \quad \text{and} \quad r^2 = x^2 + y^2 + z^2$$

$$u = \frac{1}{r} = (x^2 + y^2 + z^2)^{-1/2}$$

$$u_x = \frac{\partial u}{\partial x}$$

$$= \frac{\partial}{\partial x} (x^2 + y^2 + z^2)^{-1/2}$$

$$= -\frac{1}{2} (x^2 + y^2 + z^2)^{-1/2 - 1} \cdot 2x$$

$$= -x (x^2 + y^2 + z^2)^{-3/2}$$

$$u_{xx} = -x (-3/2) (x^2 + y^2 + z^2)^{-3/2 - 1} \cdot 2x - 1 \cdot (x^2 + y^2 + z^2)^{-3/2}$$

$$= 3x^2 (x^2 + y^2 + z^2)^{-5/2} - (x^2 + y^2 + z^2)^{-3/2} \quad \text{--- (i)}$$

Similarly,

$$u_{yy} = 3y^2 (x^2 + y^2 + z^2)^{-5/2} - (x^2 + y^2 + z^2)^{-3/2} \quad \text{--- (ii)}$$

$$u_{zz} = 3z^2 (x^2 + y^2 + z^2)^{-5/2} - (x^2 + y^2 + z^2)^{-3/2} \quad \text{--- (iii)}$$

Now, $\{ (i) + (ii) + (iii) \}$

$$u_{xx} + u_{yy} + u_{zz} = 3x^2 (x^2 + y^2 + z^2)^{-5/2} - (x^2 + y^2 + z^2)^{-3/2} +$$

$$+ 3y^2 (x^2 + y^2 + z^2)^{-5/2} - (x^2 + y^2 + z^2)^{-3/2}$$

$$+ 3z^2 (x^2 + y^2 + z^2)^{-5/2} - (x^2 + y^2 + z^2)^{-3/2}$$

$$= 3(x^2 + y^2 + z^2) (x^2 + y^2 + z^2)^{-5/2} - 3(x^2 + y^2 + z^2)^{-3/2}$$

$$= 3(x^2 + y^2 + z^2)^{-3/2} - 3(x^2 + y^2 + z^2)^{-3/2}$$

$$= 0$$

$$\therefore \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0 \quad [\text{shown}]$$

v) Hence,

$$v = -\tan^{-1} \frac{x^2 + y^2}{x - y}$$

Let,

$$\tan v = \frac{x^2 + y^2}{x + y} = f(x, y)$$

$$\begin{aligned} f(x, y) &= \frac{\partial^2 x^2 + \partial^2 y^2}{\partial x + \partial y} \\ &= \frac{\partial^2 (x^2 + y^2)}{\partial (x + y)} \end{aligned}$$

f is a homogeneous function of degree 1.

By Euler's theorem,

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = 1 \times f$$

$$\Rightarrow x \frac{\partial}{\partial x} \tan v + y \frac{\partial}{\partial y} \tan v = \tan v$$

$$\Rightarrow x \cdot \sec^2 v \frac{\partial v}{\partial x} + y \sec^2 v \frac{\partial v}{\partial y} = \tan v$$

$$\begin{aligned} \Rightarrow x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} &= \frac{\tan v}{\sec^2 v} \\ &= \frac{\frac{\sin v}{\cos v}}{\frac{1}{\cos^2 v}} \end{aligned}$$

$$= \sin u \cos u$$

$$= \frac{1}{2} (2 \sin u \cos u)$$

$$\therefore x \frac{\partial x}{\partial y} + y \frac{\partial y}{\partial x} = \frac{\sin 2u}{2}$$

[Proved]

(vi) Given,

$$x + y = 2$$

$$y = 2 - x$$

Here,

$$u = \frac{4}{x} + \frac{36}{y}$$

$$f(x) = u = \frac{4}{x} + \frac{36}{2-x}$$

$$f'(x) = \frac{du}{dx} = -\frac{4}{x^2} + \frac{36}{(2-x)^2}$$

$$f''(x) = \frac{d^2u}{dx^2} = \frac{4}{x^3} + \frac{36}{(2-x)^3}$$

For maximum or minimum,

$$-f'(x) = \frac{dv}{dx} = 0.$$

$$-\frac{4}{x^2} + \frac{36}{(2-x)^2} = 0.$$

$$\Rightarrow \frac{36}{(2-x)^2} = \frac{4}{x^2}$$

$$\Rightarrow 36x^2 = 4(4 - 4x + x^2)$$

$$\Rightarrow 32x^2 + 16x - 16 = 0.$$

$$\Rightarrow 2x^2 + x - 1 = 0.$$

$$\Rightarrow 2x^2 + 2x - x - 1 = 0.$$

$$\Rightarrow (2x-1)(x+1) = 0.$$

$$\therefore x = \frac{1}{2} \text{ or } -1.$$

$$\text{At } x = \frac{1}{2},$$

$$\begin{aligned} -f''(x) &= \frac{4}{\left(\frac{1}{2}\right)^3} + \frac{36}{\left(2 - \frac{1}{2}\right)^3} \\ &= 32 + \frac{32}{3} = \frac{128}{3} > 0. \end{aligned}$$

$$\text{At } x = -1$$

$$f''(-1) = -4 + \frac{36}{(2+1)^3}$$
$$= -\frac{8}{3} < 0.$$

at $x = -1$, maximum value —

$$f(x) = \frac{4}{x} + \frac{36}{2-x}$$

$$f(-1) = -4 + 12$$
$$= 8.$$

$x = \frac{1}{2}$, minimum value,

$$f(x) = \frac{4}{x} + \frac{36}{2-x}$$

$$f\left(\frac{1}{2}\right) = \frac{4}{\frac{1}{2}} + \frac{36}{2 - \frac{1}{2}}$$

$$= 8 + 24.$$

$$= 32$$

Example 1: A proton moves in an electric field such that its acceleration (in cms^{-2}) is $a = -\frac{20}{(1+2t)^2}$; where t is in seconds. Find the velocity as a function of time if $v = 30 \text{ cms}^{-1}$ when $t = 0$.

Solution: We have from (iv),

$$v = \int a dt$$

$$\text{So } \Rightarrow v = \int \frac{-20dt}{(1+2t)^2} \text{-----(i)}$$

Put

$$u = 1 + 2t$$

$$\Rightarrow \frac{du}{dt} = \frac{d}{dt}(1+2t)$$

$$\Rightarrow \frac{du}{dt} = 0 + 2.1$$

$$\Rightarrow \frac{du}{dt} = 2$$

$$\Rightarrow dt = \frac{du}{2}$$

$$\text{From (i), } v = \int \frac{-20dt}{(1+2t)^2} = \int \frac{-20}{u^2} \cdot \frac{du}{2} = \int \frac{-10}{u^2} du = -10 \int \frac{1}{u^2} du = -10 \int u^{-2} du$$

$$\Rightarrow v = -10 \times \frac{u^{-2+1}}{-2+1} = -10 \times \frac{u^{-1}}{-1} = 10 \times \frac{1}{u} = \frac{10}{u}$$

$$\Rightarrow v = \frac{10}{1+2t} + c \quad [\because u = 1+2t] \text{-----(ii)}$$

When $t = 0$, $v = 30$

Putting these values in (ii),

$$v = \frac{10}{1+2t} + c$$

$$\Rightarrow 30 = \frac{10}{1+2 \times 0} + c$$

$$\Rightarrow 30 = \frac{10}{1} + c$$

$$\Rightarrow 30 = 10 + c$$

$$\Rightarrow 30 - 10 = c$$

$$\Rightarrow 20 = c$$

$$\Rightarrow c = 20$$

From (ii), So the expression for velocity as a function of time is:

$$v = \frac{10}{1+2t} + 20 \text{ Cm/sec}$$

Integration

Example 3: The electric current (in mA) in a computer circuit as a function of time is $i = 0.3 - 0.2t$. What total charge passes a point in the circuit in $0.050s$?

Solution: The charge, q , is given by: $q = \int i \, dt$

$$q = \int (0.3 - 0.2t) \, dt$$

$$\Rightarrow q = \int 0.3 \, dt - \int 0.2t \, dt = 0.3 \int dt - 0.2 \int t \, dt = 0.3 \times t - 0.2 \times \frac{t^{1+1}}{1+1} + c$$

$$\Rightarrow q = 0.3 \times t - 0.2 \times \frac{t^2}{2} + c = 0.3 \times t - 0.1 \times t^2 + c$$

$$\Rightarrow q = 0.3t - 0.1t^2 + c \text{ -----(i)}$$

At $t = 0$, $q = 0$

Putting these values in (i)

$$q = 0.3t - 0.1t^2 + c$$

$$\Rightarrow 0 = 0.3 \times 0 - 0.1 \times 0^2 + c$$

$$\Rightarrow 0 = 0 + c$$

$$\Rightarrow 0 = c$$

$$\Rightarrow c = 0$$

From (i), $q = 0.3t - 0.1t^2 + c$

$$\Rightarrow q = 0.3t - 0.1t^2 + 0 [\because c = 0]$$

$$\Rightarrow q = 0.3t - 0.1t^2 \text{ -----(ii)}$$

At time $t = 0.050$

$$q = 0.3t - 0.1t^2$$

$$\Rightarrow q = 0.3 \times (0.050) - 0.1 \times (0.050)^2 = 0.015 - 0.00025 = 0.01475$$

Example: 07. $\int_0^1 e^x dx$

Here, $a = 0$, $b = 1$, $nh = b - a = 1$

We have,

$$f(x) = e^x$$

$$\therefore f(a+rh) = e^{a+rh}$$

Now,

$$\int_a^b f(x) dx = \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} hf(a+rh)$$

$$\Rightarrow \int_0^1 e^x dx = \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} hf(a+rh)$$

$$= \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} h e^{a+rh}$$

$$= h \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} e^{0+rh}$$

$$= h \lim_{h \rightarrow 0} (e^0 + e^{0+h} + e^{0+2h} + e^{0+3h} + \dots + e^{0+(n-1)h})$$

$$= h \lim_{h \rightarrow 0} (1 + e^h + e^{2h} + e^{3h} + \dots + e^{(n-1)h})$$

$$= \lim_{h \rightarrow 0} h (1 +$$

$$1 + x + x^2 + \dots + x^{n-1} = \frac{x^n - 1}{x - 1}$$

$$= \lim_{h \rightarrow 0} h(1 + eh + (eh)^2 + (eh)^3 + \dots + (eh)^{n-1})$$

$$= \lim_{h \rightarrow 0} h \frac{(eh)^n - 1}{eh - 1}$$

$$= \lim_{h \rightarrow 0} h \frac{e^{hn} - 1}{e^h - 1}$$

$$= \lim_{h \rightarrow 0} h \frac{e - 1}{e^h - 1}$$

$$= (e-1) \lim_{h \rightarrow 0} \frac{(h+0) \cdot h}{e^h - 1}$$

$$= (e-1) \lim_{h \rightarrow 0} \frac{(h+0) \cdot h^{\frac{1}{n}}}{\left[1 + h + \frac{h^2}{2!} + \frac{h^3}{3!} + \dots \right] - 1}$$

$$= (e-1) \lim_{h \rightarrow 0} \frac{h}{h \left(1 + \frac{h}{2!} + \frac{h^2}{3!} + \dots \right)}$$

$$= (e-1) \frac{1}{1 + \frac{0}{2!} + \frac{0}{3!} + \dots}$$

$$= (e-1) \cdot \frac{1}{1}$$

$$\therefore \int_0^1 e^x dx = (e-1)$$

* Example : 10. $\int_0^1 x^3 dx$

We have, $\int_a^b f(x) dx = \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} hf(a+rh)$

Given,

$$f(x) = x^3$$

$$f(a+rh) = (a+rh)^3$$

Here $a = 0$, $b = 1$

$$\Rightarrow \int_a^b f(x) dx = \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} hf(a+rh)$$

$$\Rightarrow \int_0^1 x^3 dx = \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} hf(a+rh)$$

$$= \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} h(a+rh)^3$$

$$= \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} h(0+rh)^3$$

$$= \lim_{h \rightarrow 0} h \sum_{r=0}^{n-1} (r^3 h^3)$$

$$= \lim_{h \rightarrow 0} h \times h^3 \sum_{r=0}^{n-1} r^3$$

$$= \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} h^4 r^3$$

$$= \lim_{h \rightarrow 0} h^4 \sum_{r=2}^n r^3$$

$$= \lim_{h \rightarrow 0} h^4 (1^3 + 2^3 + 3^3 + \dots + n^3)$$

$$= \lim_{h \rightarrow 0} h^4 \left\{ \frac{n(n+1)}{2} \right\}^2$$

$$= \lim_{h \rightarrow 0} h^4 \frac{n(n+1)}{2} \cdot \frac{n(n+1)}{2}$$

$$= \lim_{h \rightarrow 0} \frac{nh(nh+h)}{2} \cdot \frac{nh(nh+h)}{2}$$

$$= \lim_{h \rightarrow 0} \frac{1 \cdot (1+h)}{2} \cdot \frac{1 \cdot (1+h)}{2} \quad [nh=1]$$

$$= \frac{1 \cdot (1+0)}{2} \cdot \frac{1 \cdot (1+0)}{2}$$

$$= 1/4$$

Example 11: $\int_1^2 x^2 dx$

Solution: We have, $\int_a^b f(x) dx = \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} h f(a + rh)$

Given, $f(x) = x^2$

$$f(a + rh) = (a + rh)^2$$

Here $a = 1, b = 2$

We have, $b = a + nh$

$$\Rightarrow 2 = 1 + nh$$

$$\Rightarrow nh = 2 - 1 = 1$$

$$\Rightarrow nh = 1$$

$$\int_a^b f(x) dx = \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} h f(a + rh)$$

$$\int_1^2 x^2 dx = \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} h f(a + rh)$$

$$= \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} h (a + rh)^2 = \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} h (1 + rh)^2 = \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} h (1 + 2rh + r^2 h^2)$$

$$= \lim_{h \rightarrow 0} h \sum_{r=0}^{n-1} (1) + \lim_{h \rightarrow 0} h \sum_{r=0}^{n-1} 2rh + \lim_{h \rightarrow 0} h \sum_{r=0}^{n-1} (r^2 h^2)$$

$$= \lim_{h \rightarrow 0} h \sum_{r=1}^n (1) + \lim_{h \rightarrow 0} h \sum_{r=1}^n 2rh + \lim_{h \rightarrow 0} h \sum_{r=1}^n (r^2 h^2)$$

$$= \lim_{h \rightarrow 0} h(1 + 1 + 1 + \dots + 1) + \lim_{h \rightarrow 0} 2h^2 \sum_{r=1}^n r + \lim_{h \rightarrow 0} h^3 \sum_{r=1}^n r^2$$

$$= \lim_{h \rightarrow 0} h \times n + \lim_{h \rightarrow 0} 2h^2 (1 + 2 + 3 + \dots + n) + \lim_{h \rightarrow 0} h^3 (1^2 + 2^2 + 3^2 + \dots + n^2)$$

$$= \lim_{h \rightarrow 0} (nh) + \lim_{h \rightarrow 0} 2h^2 \frac{n(n+1)}{2} + \lim_{h \rightarrow 0} h^3 \left\{ \frac{n(n+1)(2n+1)}{6} \right\}$$

$$= \lim_{h \rightarrow 0} (nh) + \lim_{h \rightarrow 0} 2 \frac{nh(nh+h)}{2} + \lim_{h \rightarrow 0} \left\{ \frac{nh(nh+h)(2nh+h)}{6} \right\}$$

$$= \lim_{h \rightarrow 0} (1) + \lim_{h \rightarrow 0} 2 \frac{1 \cdot (1+h)}{2} + \lim_{h \rightarrow 0} \left\{ \frac{1 \cdot (1+h)(2 \cdot 1+h)}{6} \right\}$$

$$= (1) + 2 \frac{1 \cdot (1+0)}{2} + \left\{ \frac{1 \cdot (1+0)(2 \cdot 1+0)}{6} \right\}$$

$$= (1) + 1 + \left\{ \frac{(1)(2)}{6} \right\} = (1) + 1 + \frac{1}{3} = \frac{7}{3} \text{ Answer}$$

Example: 35. Find the area bounded by the curves by $y = x^2$, $y = 2 - x$ and $y = 1$.

We need to solve $y = x^2$ for x .

Given,

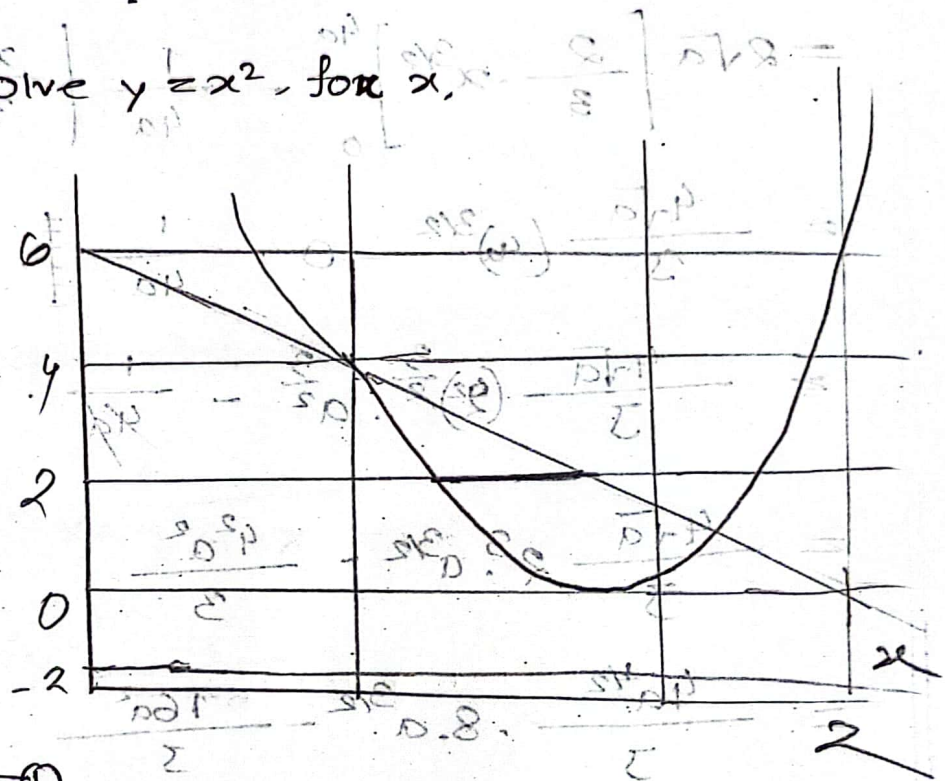
$$y = x^2$$

$$\Rightarrow y^{1/2} = (x^2)^{1/2}$$

$$\Rightarrow y^{1/2} = x$$

$$\Rightarrow \pm \sqrt{y} = x$$

$$\Rightarrow x = \pm \sqrt{y} \quad \text{--- (i)}$$



Again,

$$y = 2 - x$$

$$\Rightarrow x = 2 - y \quad \text{--- (ii)}$$

We need the left hand portion, so $x = -\sqrt{y}$

Notice that,

$\Rightarrow x = 2 - y$ is to the right of $x = -\sqrt{y}$, so we

choose,

$$x_2 = 2 - y \quad \text{--- (iii)}$$

$$x_1 = -\sqrt{y} \quad \text{--- (iv)}$$

The intersection of the graphs occurs at

$$2 - y = -\sqrt{y}$$

$$\Rightarrow (2 - y)^2 = (-\sqrt{y})^2$$

$$\Rightarrow 4 - 4y + y^2 = y$$

$$\Rightarrow y^2 - 5y + 4 = 0$$

$$\Rightarrow y(y - 4) - 1(y - 4) = 0$$

$$\Rightarrow (y - 4)(y - 1) = 0$$

$\therefore y = 4, y = 1$

putting the value of $y = 4$ in (iii) we get

$$x_2 = 2 - y$$

$$\Rightarrow x_2 = 2 - 4 = -2$$

putting the value of $y = 1$ in (ii) we get

$$x_1 = -\sqrt{y}$$

$$\Rightarrow x_1 = -\sqrt{1} = -1$$

Hence the point of intersection $(-2, 4)$

Again, putting the value of $y = 1$ in (iii) we get,

$$x_2 = 2 - y$$

$$\Rightarrow x_2 = 2 - 1 = 1$$

And putting the value of $y = 1$ in (iv) we

$$x_1 = -\sqrt{y}$$

$$= -\sqrt{1} = -1$$

intersection $(-1, 1)$

∴ $(-2, 4)$ and $(-1, 1)$ between the

curves $y = x^2$ and $y = 2 - x$ to solve with putting

$$c = 1 \text{ and } d = 4$$

$$A = \int_a^b (x_2 - x_1) dy$$

$$A = \int_c^d (x_2 - x_1) dy$$

$$= \int_1^4 [2 - y] - [-\sqrt{y}] dy$$

$$= \int_1^4 (2 - y + \sqrt{y}) dy = \left[2y - \frac{y^2}{2} + \frac{2}{3} y^{3/2} \right]_1^4$$

$$2 \frac{16}{3} - \frac{13}{6}$$

$$= \frac{19}{6} \text{ sq. unit.}$$

Example 50: $\int \frac{3x dx}{x^2 - x - 2}$

Solution:

01. Here denominator $x^2 - x - 2$

$$\therefore \frac{d}{dx}(x^2 - x - 2) = 2x - 1$$

02. $2x - 1$ কে প্রথমে লবে (Numerator) লিখে ফেলতে হবে।

03. কিন্তু উপরে (Numerator) আছে $3x$, এই $3x$ এর সাথে balance করার জন্য $2x - 1$ এর সাথে $\frac{3}{2}$ দ্বারা গুন করতে হবে। মনে রাখতে হবে x এর সহগের সাথে যত গুন থাকবে তত দ্বারা ভাগ করতে হবে। এখানে $2x - 1$ এ x এর সাথে ২ গুন আছে এজন্য ২ দ্বারা ভাগ করতে হবে।

04. এর পর $3x$ সাথে balance করার জন্য $\frac{3}{2}$ যোগ করতে হবে।

$$05. \text{ অর্থাৎ } 3x = \frac{3}{2}(2x - 1) + \frac{3}{2} = \frac{3}{2} \times 2x - \frac{3}{2} + \frac{3}{2} = 3x - \frac{3}{2} + \frac{3}{2} = 3x$$

$$\therefore \int \frac{3x dx}{x^2 - x - 2} = \int \frac{\frac{3}{2}(2x - 1) + \frac{3}{2}}{x^2 - x - 2} dx \quad [\text{Method \# 04}]$$

$$= \frac{3}{2} \int \frac{2x - 1}{x^2 - x - 2} dx + \frac{3}{2} \int \frac{dx}{x^2 - x - 2}$$

[মনে রাখুন, এখানে দুটি integration এর প্রথমটি Method # 03 এবং পরেরটি Method # 02 অনুসারে]

$$= \frac{3}{2} \log(x^2 - x - 2) + \frac{3}{2} \int \frac{dx}{x^2 - x - 2} \left[\int \frac{f'(x)}{f(x)} dx = \ln f(x) + c \text{ \& Method-03} \right]$$

[নিচের ফাংশনকে ডিফারেন্সিয়েট করলে যদি উপরের ফাংশন পাওয়া যায় তাহলে তার ইন্টিগ্রেশন হল লগ অফ নিচের ফাংশন :]

$$= \frac{3}{2} \log(x^2 - x - 2) + \frac{3}{2} \int \frac{dx}{x^2 - 2x \cdot \frac{1}{2} + \left(\frac{1}{2}\right)^2 - \frac{1}{4} - 2} \quad [\text{Method \# 02}]$$

$$= \frac{3}{2} \log(x^2 - x - 2) + \frac{3}{2} \int \frac{dx}{\left(x - \frac{1}{2}\right)^2 - \frac{9}{4}}$$

$$= \frac{3}{2} \log(x^2 - x - 2) + \frac{3}{2} \int \frac{dx}{\left(x - \frac{1}{2}\right)^2 - \left(\frac{3}{2}\right)^2}$$

$$= \frac{3}{2} \log(x^2 - x - 2) + \frac{3}{2} \cdot \frac{1}{\frac{3}{2}} \log \frac{x - \frac{1}{2} - \frac{3}{2}}{x - \frac{1}{2} + \frac{3}{2}} \quad [\text{Formula \#27}]$$

$$= \frac{3}{2} \log(x^2 - x - 2) + \frac{1}{2} \log \frac{x - 2}{x + 1} \quad \text{Answer}$$

Example: 56.

$$\frac{dx}{(2x+1)\sqrt{4x+3}}$$

$$= \int \frac{\frac{1}{2} z dz}{\left(2x \frac{z^2-3}{4} + 1\right) \sqrt{z^2}}$$

$$= \frac{1}{2} \int \frac{z dz}{\left(\frac{z^2-3}{2} + 1\right) z}$$

$$= \frac{1}{2} \int \frac{z dz}{\left(\frac{z^2-3+2}{2}\right) z}$$

$$= \frac{1}{2} \int \frac{z dz}{(z^2-1) z}$$

Let,

$$z^2 = 4x+3$$

$$\text{or, } 2z dz = 4 dx$$

$$\text{or, } z dz = 2 dx$$

$$\text{or, } dx = \frac{1}{2} z dz$$

Again,

$$z^2 = 4x+3$$

$$\text{or, } x = \frac{z^2-3}{4}$$

$$= \int \frac{dz}{z^2-1}$$

$$= \frac{1}{2.1} \log \frac{z-1}{z+1} \quad \frac{1+\sqrt{z}}{1-\sqrt{z}} \times \frac{1}{p.} =$$

$$= \frac{1}{2} \log \frac{\sqrt{4x+3} - 1}{\sqrt{4x+3} + 1} \quad \frac{1+\sqrt{z}}{1-\sqrt{z}} \times \frac{1}{p.} =$$

Example: 57.

$$\int \frac{dx}{(2+x)\sqrt{1+x}}$$

$$= \int \frac{2z dz}{(2+z^2-1)\sqrt{z^2}}$$

$$= 2 \int \frac{dz}{z^2+1}$$

$$= 2 \tan^{-1} \frac{z}{1}$$

$$= 2 \tan^{-1} \sqrt{1+x}$$

Let,

$$z^2 = 1+x$$

$$\text{or, } 2z dz = dx$$

Again,

$$z^2 = 1+x$$

$$\text{or, } x = z^2 - 1$$

Example 58:

$$\begin{aligned} & \int \frac{dx}{(2x+3)\sqrt{x^2+3x+2}} \\ \therefore \int & \frac{dx}{(2x+3)\sqrt{x^2+3x+2}} \\ & -\frac{1}{2z^2} dz \\ = \int & \frac{\frac{1}{z} \sqrt{\frac{1}{4}(\frac{1}{z}-3)^2 + 3 \cdot \frac{1}{2}(\frac{1}{z}-3) + 2}}{-\frac{1}{z^2} dz} \\ = \frac{1}{2} \int & \frac{-\frac{1}{z^2} dz}{\frac{1}{z} \sqrt{\frac{1}{4}(\frac{1}{z}-3)^2 + 3 \cdot \frac{1}{2}(\frac{1}{z}-3) + 2}} \end{aligned}$$

$$\text{Let, } 2x+3 = \frac{1}{z}$$

$$\Rightarrow 2dx = -\frac{1}{z^2} dz$$

$$\Rightarrow dx = -\frac{1}{2z^2} dz$$

$$\text{Again, } 2x+3 = \frac{1}{z}$$

$$\Rightarrow 2x = \frac{1}{z} - 3$$

$$\therefore x = \frac{1}{2} \left(\frac{1}{z} - 3 \right) = \frac{1-3z}{2z}$$

$$\begin{aligned} & = \frac{1}{2} \int \frac{-zdz}{z^2 \sqrt{\frac{1-6z+9z^2}{4z^2} + \frac{3-9z}{2z} + 2}} \\ & = \frac{1}{2} \int \frac{-dz}{z \sqrt{\frac{1-6z+9z^2+6z-18z^2+8z^2}{4z^2}}} = \frac{1}{2} \int \frac{-dz}{z \sqrt{\frac{1-z^2}{4z^2}}} = \frac{1}{2} \int \frac{-dz}{z \cdot \frac{1}{2z} \sqrt{1-z^2}} \\ & = \frac{2}{2} \int \frac{-dz}{\sqrt{1-z^2}} = \int \frac{-dz}{\sqrt{1-z^2}} = -\sin^{-1} z \\ & = -\sin^{-1} \left(\frac{1}{2x+3} \right) \left[\because 2x+3 = \frac{1}{z}; \therefore z = \frac{1}{2x+3} \right] \text{ [Formula \#11]} \end{aligned}$$

Example: 6.1. $\int \frac{dx}{x^2 + x - 2}$ (i) if $L < R$ partial fraction

(S+1)B + (R-1)A = 1

The integrand is a proper rational function that can be written as,

$$\frac{1}{x^2 + x - 2} = \frac{1}{x^2 + 2x - x - 2} = \frac{1}{x(x+2) - 1(x+2)} = \frac{1}{(x+2)(x-1)} \quad \text{--- (i)}$$

Let, $\frac{1}{(x+2)(x-1)} = \frac{A}{x+2} + \frac{B}{x-1}$ (ii)

$$\frac{1}{(x+2)(x-1)} = \frac{A}{x+2} + \frac{B}{x-1}$$

Multiplying by denominator $(x+2)(x-1)$.

$$\frac{1}{(x+2)(x-1)} \times (x+2)(x-1) = \frac{A}{x+2} \times (x+2)(x-1) + \frac{B}{x-1} \times (x+2)(x-1)$$

$$\Rightarrow 1 = A(x-1) + B(x+2) \quad \text{--- (iii)}$$

Let, $x+2 = 0$

∴, $x = -2$

putting $x = -2$ in (iii)

$$1 = A(-2-1) + B(-2+2) \quad \text{--- ∴, } A = -1/3$$

Again, $x - 1 = 0$

$\therefore x = 1$

putting $x = 1$ in (ii)

$$1 = A(1-1) + B(1+2)$$

$\therefore B = \frac{1}{3}$

putting A & B values in (i)

$$\frac{1}{(x+2)(x-1)} = \frac{-1/3}{x+2} + \frac{1/3}{x-1}$$

$$\therefore \frac{1}{x^2+x-2} = \frac{-1/3}{x+2} + \frac{1/3}{x-1}$$

$$\int \frac{dx}{x^2+x-2} = \int \frac{-1/3}{x+2} dx + \int \frac{1/3}{x-1} dx$$

$$= -\frac{1}{3} \int \frac{1}{x+2} dx + \frac{1}{3} \int \frac{1}{x-1} dx$$

$$= -\frac{1}{3} \ln(x+2) + \frac{1}{3} \ln(x-1)$$

$$a \cos x + b \sin x = I \left(\frac{\text{Differentiate of Denominator}}{m(\text{Denominator})} \right) +$$

✓ Example : 68. $\int \frac{3 \cos x + 4 \sin x}{4 \cos x + 5 \sin x} dx$

Let,

$$I = \int \frac{3 \cos x + 4 \sin x}{4 \cos x + 5 \sin x} dx$$

Again,

$$3 \cos x + 4 \sin x = I \left\{ \frac{d}{dx} (4 \cos x + 5 \sin x) \right\} + m (4 \cos x + 5 \sin x) \quad \text{--- (i)}$$

$$\Rightarrow 3 \cos x + 4 \sin x = I (-4 \sin x + 5 \cos x) + m (4 \cos x + 5 \sin x)$$

$$\Rightarrow 3 \cos x + 4 \sin x = -4I \sin x + 5I \cos x + 4m \cos x + 5m \sin x$$

$$\Rightarrow 3 \cos x + 4 \sin x = \cos x (5I + 4m) + \sin x (-4I + 5m)$$

Equating the co-efficient of $\cos x$ and $\sin x$ we get

$$\therefore 3 \cos x = \cos x (5I + 4m)$$

$$\Rightarrow 3 = 5I + 4m \quad \text{--- (ii)}$$

$$\therefore 4 \sin x = (-4I + 5m) \sin x$$

$$\Rightarrow 4 = -4I + 5m \quad \text{--- (iii)}$$

$$\textcircled{ii} \times 4 + \textcircled{iii} \times 5 \rightarrow (4 \cos x + 5 \sin x)$$

$$12 + 20 = 20I + 16m - 20I + 25m$$

$$\therefore m = \frac{32}{41}$$

$$m = \frac{32}{41} \text{ in } \textcircled{ii} \rightarrow (4 \cos x + 5 \sin x)$$

$$3 = 5I + 4 \times \frac{32}{41}$$

$$\therefore 5I = -\frac{5}{41}$$

$$\therefore I = -\frac{1}{41}$$

putting the value of I, m in $\textcircled{i}$

$$3 \cos x + 4 \sin x = \frac{1}{41} \left\{ \frac{d}{dx} (4 \cos x + 5 \sin x) \right\} + m(4 \cos x + 5 \sin x)$$

$$\therefore 3 \cos x + 4 \sin x = -\frac{1}{41} (-4 \sin x + 5 \cos x) + \frac{32}{41} (4 \cos x + 5 \sin x) \quad \textcircled{iv}$$

$$\therefore I = \int \frac{3 \cos x + 4 \sin x}{4 \cos x + 5 \sin x} dx$$

$$= \int \frac{-\frac{1}{41} (-4 \sin x + 5 \cos x) + \frac{32}{41} (4 \cos x + 5 \sin x)}{4 \cos x + 5 \sin x} dx$$

$$= \int \left\{ -\frac{1}{41} \frac{(-4\sin x + 5\cos x)}{4\cos x + 5\sin x} + \frac{32}{41} \frac{4\cos x + 5\sin x}{4\cos x + 5\sin x} \right\} dx$$

$$= \int \left\{ -\frac{1}{41} \times \frac{(-4\sin x + 5\cos x)}{4\cos x + 5\sin x} dx + \frac{32}{41} dx \right\}$$

$$= -\frac{1}{41} \ln|4\cos x + 5\sin x| + \frac{32}{41} x + C$$

Example: 62. $\int \frac{3x-1}{(x+4)^2} dx$

Let,

$$\frac{3x-1}{(x+4)^2} = \frac{A}{(x+4)} + \frac{B}{(x+4)^2} \quad \text{--- (1)}$$

Multiplying (1) by the denominator $(x+4)^2$

$$\frac{3x-1}{(x+4)^2} \times (x+4)^2 = \frac{A}{(x+4)} \times (x+4)^2 + \frac{B}{(x+4)^2} \times (x+4)^2$$

$$\Rightarrow 3x-1 = A(x+4) + B \quad \text{--- (2)}$$

Let,

$$x+4=0$$

$$\therefore x = -4$$

Putting $x = -4$ in (2) $\rightarrow$

$$3(-4) - 1 = A(-4+4) + B$$

$$\therefore B = -13$$

Equation (2) can be written as,

$$(3x-1) = A(x+4) + B$$

$$\therefore (3x-1) = Ax + 4A + B$$

$$\therefore 3x - 1 = Ax + 4A + B \quad \text{--- (3)}$$

From (ii) we get, $\frac{3x-1}{(x+4)^2}$ Example: 63

$$3 = A$$

$$-1 = 4A + B$$

putting the value of A and B in (i),

$$\frac{3x-1}{(x+4)^2} = \frac{A}{x+4} + \frac{B}{(x+4)^2}$$

$$= \frac{3}{(x+4)} + \frac{-13}{(x+4)^2}$$

$$\therefore \int \frac{3x-1}{(x+4)^2} dx = \int \frac{3}{(x+4)} dx + \int \frac{-13}{(x+4)^2} dx$$

$$= \int \frac{3}{z} dz + \int \frac{-13}{z^2} dz$$

$$= \int \frac{3}{z} dz - 13 \int \frac{1}{z^2} dz$$

$$= 3 \int \frac{1}{z} dz - 13 \int z^{-2} dz$$

$$= 3 \ln z - 13 \frac{z^{-2+1}}{-2+1}$$

$$= 3 \ln z + 13 z^{-1}$$

$$= 3 \ln z + 13 \frac{1}{z}$$

$$= 3 \ln(x+4) + \frac{13}{(x+4)}$$

Example: 70. $\int x e^x dx$

We have,

$$\int u dv = uv - \int v du \quad \text{--- (1)}$$

Let,

$$u = x$$

$$\Rightarrow \frac{du}{dx} = 1$$

$$\Rightarrow du = dx$$

Let,

$$dv = e^x dx$$

$$\Rightarrow \int dv = \int e^x dx$$

$$\Rightarrow v = e^x$$

Putting the values of u, dv, v, du in (1) $\rightarrow$

$$\int u dv = uv - \int v du$$

$$\Rightarrow \int x e^x dx = x e^x - \int e^x dx$$

$$\Rightarrow x e^x dx = x e^x - e^x + C$$

Example: 76. $\int e^x \{f(x) + f'(x)\} dx = e^x f(x) + C$

$$\int e^x (\sin x + \cos x) dx$$

$$= e^x \sin x + C$$

Example : 77. $\int e^x (\sin x - \cos x) dx$

$$= \int e^x (-\cos x) \sin x + (-\cos x) dx$$

$$= \int e^x (-\cos x + \sin x) dx$$

$$= e^x (-\cos x) + c$$

Here, $f(x) = -\cos x$; $f'(x) = -(-\sin x) = \sin x$

Example 72: Find $\int \tan^{-1} x dx$

Solution:

We have, $\int u dv = uv - \int v du$ -----(i)

$$\text{Given, } \int \underbrace{\tan^{-1} x}_u \underbrace{dx}_{dv}$$

Let

$$u = \tan^{-1} x$$

$$\frac{du}{dx} = \frac{1}{1+x^2}$$

$$du = \frac{dx}{1+x^2}$$

Let

$$dv = dx$$

$$\int dv = \int dx$$

$$\therefore v = x$$

Putting the values of u , dv , v , du in (i)

$$\int u dv = uv - \int v du$$

$$\int \underbrace{\tan^{-1} x}_u \underbrace{dx}_{dv} = \underbrace{(\tan^{-1} x)}_u \underbrace{x}_v - \int \underbrace{x}_v \underbrace{\frac{dx}{1+x^2}}_{du}$$

$$= x \tan^{-1} x - \int \frac{x}{1+x^2} dx = x \tan^{-1} x - \frac{1}{2} \int \frac{2x}{1+x^2} dx.$$

$$= x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) \text{ [Formula\#63] Answer}$$

* Example: 83. Prove that, $I_n = \int_0^{\pi/2} \sin^n x dx = \frac{n-1}{n} I_{n-2}$

$$I_n = \int_0^{\pi/2} \sin^n x dx$$

$$= \left[-\frac{1}{n} \sin^{n-1} x \cos x \right]_0^{\pi/2} + \frac{n-1}{n} \int_0^{\pi/2} \sin^{n-2} x dx$$

$$= \left[-\frac{1}{n} \sin^{n-1} \left(\frac{\pi}{2} \right) \cos \frac{\pi}{2} + \frac{1}{n} \sin^{n-1}(0) \cos 0 \right] + \frac{n-1}{n} \int_0^{\pi/2} \sin^{n-2} x dx$$

$$= \left[-\frac{1}{n} \cdot 1 \cdot 0 + 0 \cdot 1 \right] + \frac{n-1}{n} \int_0^{\pi/2} \sin^{n-2} x dx$$

$$= \frac{n-1}{n} \int_0^{\pi/2} \sin^{n-2} x dx$$

$$= \frac{n-1}{n} I_{n-2} \quad \left[\because I_n = \int_0^{\pi/2} \sin^n x dx \therefore I_{n-2} = \int_0^{\pi/2} \sin^{n-2} x dx \right]$$

Proved

* Example : 84, Prove that, $I_n = \int_0^{\pi/2} \cos^n x dx = \frac{n-1}{n} I_{n-2}$

$$I_n = \int_0^{\pi/2} \cos^n x dx$$

$$= \left[\frac{1}{n} \cos^{n-1} x \sin x \right]_0^{\pi/2} + \frac{n-1}{n} \int_0^{\pi/2} \cos^{n-2} x dx$$

$$= \left[\frac{1}{n} \cos^{n-1} \left(\frac{\pi}{2} \right) \sin \frac{\pi}{2} - \frac{1}{n} \cos^{n-1}(0) \sin 0 \right] + \frac{n-1}{n} \int_0^{\pi/2} \cos^{n-2} x dx$$

$$= \left[\frac{1}{n} \cdot 0 \cdot 1 - \frac{1}{n} \cdot 1 \cdot 0 \right] + \frac{n-1}{n} \int_0^{\pi/2} \cos^{n-2} x dx$$

$$= \frac{n-1}{n} I_{n-2}$$

Proved

$$\left[\because I_n = \int_0^{\pi/2} \cos^n x dx \right]$$

$$\left[\therefore I_{n-2} = \int_0^{\pi/2} \cos^{n-2} x dx \right]$$

Example 93: If $I_n = \int_0^{\pi/4} \tan^n \theta d\theta$ and $n > 1$, Prove that $n(I_{n-1} + I_{n+1}) = 1$

Solution: Given, $I_n = \int_0^{\pi/4} \tan^n \theta d\theta$ -----(i)

Putting $n = n + 1$ in (i)

$$I_{n+1} = \int_0^{\pi/4} \tan^{n+1} \theta d\theta$$

$$I_{n+1} = \int_0^{\pi/4} \tan^{n-1} \theta \cdot \tan^2 \theta d\theta$$

$$I_{n+1} = \int_0^{\pi/4} \tan^{n-1} \theta (\sec^2 \theta - 1) d\theta \quad [\because \sec^2 \theta - 1 = \tan^2 \theta]$$

$$I_{n+1} = \int_0^{\pi/4} \tan^{n-1} \theta \sec^2 \theta d\theta - \int_0^{\pi/4} \tan^{n-1} \theta d\theta$$
 -----(ii)

Now, $\int \tan^{n-1} \theta \sec^2 \theta d\theta$ -----(iii)

From (iii)

$$\int \tan^{n-1} \theta \sec^2 \theta d\theta$$

$$= \int z^{n-1} dz$$

$$= \frac{z^{n-1+1}}{n-1+1} = \frac{z^n}{n}$$

$$= \frac{\tan^n \theta}{n} \quad [z = \tan \theta]$$

Let $z = \tan \theta$

$$\frac{dz}{d\theta} = \sec^2 \theta$$

$$dz = \sec^2 \theta d\theta$$

From (ii)

$$I_{n+1} = \int_0^{\pi/4} \tan^{n-1} \theta \sec^2 \theta d\theta - \int_0^{\pi/4} \tan^{n-1} \theta d\theta$$

$$I_{n+1} = \left[\frac{\tan^n \theta}{n} \right]_0^{\pi/4} - \int_0^{\pi/4} \tan^{n-1} \theta d\theta$$

$$I_{n+1} = \left[\frac{\tan^n \theta}{n} \right]_0^{\pi/4} - I_{n-1} \quad [\because I_n = \int_0^{\pi/4} \tan^n \theta d\theta \therefore I_{n-1} = \int_0^{\pi/4} \tan^{n-1} \theta d\theta]$$

$$I_{n+1} + I_{n-1} = \left[\frac{\tan^n \theta}{n} \right]_0^{\pi/4}$$

$$I_{n+1} + I_{n-1} = \left[\frac{\tan^n \frac{\pi}{4}}{n} - \frac{\tan^n 0}{n} \right] = \left[\frac{(\tan \frac{\pi}{4})^n}{n} - \frac{(\tan 0)^n}{n} \right]$$

$$I_{n+1} + I_{n-1} = \left[\frac{1^n}{n} - \frac{0}{n} \right] = \left[\frac{1}{n} \right]$$

$$I_{n+1} + I_{n-1} = \frac{1}{n}$$

$$n(I_{n+1} + I_{n-1}) = 1 \text{ Proved}$$

$$* \int_0^a f(x) dx = \int_0^a f(a-x) dx$$

Example: 99. Show that $\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx = \frac{\pi}{4}$

Let,

$$I = \int_0^{\pi/2} \frac{\sin(\pi/2 - x)}{\sin(\pi/2 - x) + \cos(\pi/2 - x)} dx \quad \rightarrow I = \int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx \quad \text{--- (1)}$$

$$I = \int_0^{\pi/2} \frac{\cos x}{\cos x + \sin x} dx \quad \text{--- (2)}$$

From (1) + (2) $\rightarrow$

$$2I = \int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx + \int_0^{\pi/2} \frac{\cos x}{\cos x + \sin x} dx$$

$$= \int_0^{\pi/2} \frac{(\sin x + \cos x)}{(\sin x + \cos x)} dx$$

$$= \int_0^{\pi/2} dx$$

$$= [x]_0^{\pi/2}$$

$$= \left[\frac{\pi}{2} - 0 \right]$$

$$= \pi/2$$

$$\Rightarrow 2I = \pi/2$$

$$\Rightarrow I = \pi/4 \quad \text{shaded}$$

Example : 100. Let,

$$I = \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx \quad \text{--- (i)}$$

$$\text{also } I = \int_0^{\pi} \frac{(\pi - x) \sin(\pi - x)}{1 + \cos^2(\pi - x)} dx$$

$\xrightarrow{-\cos x}$

$$I = \int_0^{\pi} \frac{(\pi - x) \sin x}{1 + \cos^2 x} dx \quad \text{--- (ii)}$$

from (i) + (ii) $\rightarrow$

$$2I = \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx + \int_0^{\pi} \frac{(\pi - x) \sin x}{1 + \cos^2 x} dx$$

$$= \int_0^{\pi} \frac{x \sin x + (\pi - x) \sin x}{1 + \cos^2 x} dx$$

$$= \int_0^{\pi} \frac{\pi \sin x}{1 + \cos^2 x} dx \quad \text{--- (iii)}$$

P.T.O

The gamma and Beta function

Example: 12.1. Prove that, $\Gamma(1) = 1$

We have,

$$\int_0^{\infty} x^{n-1} e^{-x} dx \quad \text{--- (1)}$$

Put $n=1$ in (1)

$$\Gamma(1) = \int_0^{\infty} x^{1-1} e^{-x} dx$$

$$\Rightarrow \Gamma(1) = \int_0^{\infty} 1 \cdot e^{-x} dx \quad [\because x^0 = 1]$$

$$\Rightarrow \Gamma(1) = \int_0^{\infty} e^{-x} dx \Rightarrow \left[\frac{e^{-x}}{-1} \right]_0^{\infty}$$

$$= - \left[e^{-x} \right]_0^{\infty}$$

$$= - \left[e^{-\infty} - e^{-0} \right]$$

$$= - \left[0 - 1 \right]$$

$$= - [0 - 1]$$

$$\frac{1}{0} = 0$$

$$= 1$$

proved

Example: 125. $\int_0^{\infty} x^7 e^{-x} dx$

We can write the above function,

$$\int_0^{\infty} x^7 e^{-x} dx = \int_0^{\infty} x^{8-1} e^{-x} dx \quad \text{--- (1)}$$

We have,

$$\Gamma(n) = \int_0^{\infty} x^{n-1} e^{-x} dx$$

$$\therefore \Gamma(8) = \int_0^{\infty} x^{8-1} e^{-x} dx \quad \text{--- (2)}$$

from (1) $\rightarrow$

$$\therefore \int_0^{\infty} x^7 e^{-x} dx = \int_0^{\infty} x^{8-1} e^{-x} dx$$

$$= \Gamma(8)$$

$$= (8-1)!$$

$$= 7!$$

$$= 5040.$$

$$[\Gamma(n) = (n-1)!]$$

Example III: Show that $\int_0^{\pi/2} \frac{(\sin x)^{3/2}}{(\sin x)^{3/2} + (\cos x)^{3/2}} dx = \frac{1}{4} \pi$

Solution: Let $I = \int_0^{\pi/2} \frac{(\sin x)^{3/2}}{(\sin x)^{3/2} + (\cos x)^{3/2}} dx$ -----(i)

$$I = \int_0^{\pi/2} \frac{\left\{ \sin \left(\frac{\pi}{2} - x \right) \right\}^{3/2}}{\left\{ \sin \left(\frac{\pi}{2} - x \right) \right\}^{3/2} + \left\{ \cos \left(\frac{\pi}{2} - x \right) \right\}^{3/2}} dx$$

$$\left[\int_0^a f(x) dx = \int_0^a f(a-x) dx \right]$$

$$I = \int_0^{\pi/2} \frac{\left\{ \sin \left(1 \cdot \frac{\pi}{2} - x \right) \right\}^{3/2}}{\left\{ \sin \left(1 \cdot \frac{\pi}{2} - x \right) \right\}^{3/2} + \left\{ \cos \left(1 \cdot \frac{\pi}{2} - x \right) \right\}^{3/2}} dx$$

$$I = \int_0^{\pi/2} \frac{(\cos x)^{3/2}}{(\cos x)^{3/2} + (\sin x)^{3/2}} dx$$
 -----(ii)

(i)+(ii)

$$I + I = \int_0^{\pi/2} \frac{(\sin x)^{3/2}}{(\cos x)^{3/2} + (\sin x)^{3/2}} dx + \int_0^{\pi/2} \frac{(\cos x)^{3/2}}{(\cos x)^{3/2} + (\sin x)^{3/2}} dx$$

$$2I = \int_0^{\pi/2} \frac{(\sin x)^{3/2} + (\cos x)^{3/2}}{(\cos x)^{3/2} + (\sin x)^{3/2}} dx = \int_0^{\pi/2} \left[\frac{(\sin x)^{3/2} + (\cos x)^{3/2}}{(\sin x)^{3/2} + (\cos x)^{3/2}} \right] dx$$

$$2I = \int_0^{\pi/2} dx = \left[\frac{\pi}{2} - 0 \right] = \frac{\pi}{2}$$

$$I = \frac{\pi}{4} \text{ Proved}$$

Example 75: $\int x \log x dx$

$$= \int \log x \cdot x dx$$

$$= \log x \int x dx - \int \left\{ \frac{d}{dx} (\log x) \int x dx \right\} dx \quad [\because \int u v dx = u \int v dx - \int \left\{ \frac{d}{dx} (u) \int v dx \right\} dx]$$

$$= \log x \cdot \frac{x^2}{2} - \int \left\{ \frac{1}{x} \cdot \frac{x^2}{2} \right\} dx = \log x \cdot \frac{x^2}{2} - \frac{1}{2} \int x dx$$

$$= \frac{x^2}{2} \log x - \frac{1}{2} \cdot \frac{x^2}{2} \quad \left[\because \int x^n dx = \frac{x^{n+1}}{n+1} + c \right]$$

$$= \frac{x^2}{2} \log x - \frac{1}{4} x^2 \text{ Answer}$$

Example: 132. prove that $\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$

We have,

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx \quad \text{--- (1)}$$

put, $x = \sin^2 \theta$ in (1) $\rightarrow$

$$\Rightarrow \frac{dx}{d\theta} = \frac{d}{d\theta} (\sin^2 \theta)$$

$$\Rightarrow \frac{dx}{d\theta} = 2 \sin \theta \frac{d}{d\theta} (\sin \theta)$$

$$\Rightarrow \frac{dx}{d\theta} = 2 \sin \theta \cos \theta$$

$$\Rightarrow dx = 2 \sin \theta \cos \theta d\theta$$

Again,

$$\sin^2 \theta = x$$

$$\Rightarrow \sin \theta = \sqrt{x}$$

$$\therefore \theta = \sin^{-1} \sqrt{x}$$

x	0	1
$x = \sin^2 \theta$	$\theta = \sin^{-1} \sqrt{x}$	$\theta = \sin^{-1} \sqrt{1}$
$\sin \theta = \sqrt{x}$	$\theta = \sin^{-1} 0$	$= \sin^{-1} 1$
$\therefore \theta = \sin^{-1} \sqrt{x}$	$\theta = \sin^{-1} \sin \theta$	$= \sin^{-1} \sin \frac{\pi}{2}$
	$\theta = 0$	$= \pi/2$

From (1) $\rightarrow$

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$= \int_0^{\pi/2} (\sin^2 \theta)^{m-1} (1 - \sin^2 \theta)^{n-1} \times 2 \sin \theta \cos \theta d\theta$$

$$= 2 \int_0^{\pi/2} (\sin \theta)^{2m-2} \cdot \sin \theta \cdot (\cos \theta)^{2n-2} \cdot \cos \theta d\theta$$

$$= 2 \int_0^{\pi/2} (\sin \theta)^{2m-2+1} \cdot (\cos \theta)^{2n-2+1} d\theta$$

$$= 2 \int_0^{\pi/2} (\sin \theta)^{2m-1} \cdot (\cos \theta)^{2n-1} d\theta$$

Proved

Example: 142. Show that,

$$\int_0^{\pi/2} \cos^8 x \sin^6 x dx = \frac{5\pi}{4096}$$

Let, $I = \int_0^{\pi/2} \cos^8 x \sin^6 x dx$

$$\beta(m, n) = \frac{\Gamma\left(\frac{m+1}{2}\right) \Gamma\left(\frac{n+1}{2}\right)}{2 \Gamma\left(\frac{m+n+2}{2}\right)}$$

$$\beta(8, 6) = \frac{\Gamma\left(\frac{8+1}{2}\right) \Gamma\left(\frac{6+1}{2}\right)}{2 \Gamma\left(\frac{8+6+2}{2}\right)}$$

$$= \frac{\Gamma\left(\frac{9}{2}\right) \Gamma\left(\frac{7}{2}\right)}{2 \Gamma(8)}$$

$$= \frac{\Gamma\left(\frac{7}{2} + 1\right) \Gamma\left(\frac{5}{2} + 1\right)}{2 \times (8-1)!}$$

$$= \frac{\frac{7}{2} \Gamma\left(\frac{7}{2}\right) \cdot \frac{5}{2} \Gamma\left(\frac{5}{2}\right)}{2 \times 7!}$$

$$= \frac{\frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \Gamma\left(\frac{1}{2}\right) \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \Gamma\left(\frac{1}{2}\right)}{2 \times 7!}$$

$$= \frac{7 \cdot 5 \cdot 3 \cdot 5 \cdot 3 \cdot 1 \cdot \sqrt{\pi} \cdot \sqrt{\pi}}{2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 7!}$$

$$= \frac{5\pi}{4096}$$

Proved

Example: 148. Show that, $\int_0^{\pi} x \cos^4 x \, dx = \frac{3\pi^2}{16}$

$$\text{Let, } I = \int_0^{\pi} x \cos^4 x \, dx$$

$$= \int_0^{\pi} (\pi - x) \cos^4 (\pi - x) \, dx$$

$$= \int_0^{\pi} \pi \cos^4 (\pi - x) \, dx - \int_0^{\pi} x \cos^4 (\pi - x) \, dx$$

$$= \int_0^{\pi} \pi \{ \cos (\pi - x) \}^4 \, dx - \int_0^{\pi} x \{ \cos (\pi - x) \}^4 \, dx$$

$$= \int_0^{\pi} \pi (-\cos x)^4 \, dx - \int_0^{\pi} x (-\cos x)^4 \, dx$$

$$= \int_0^{\pi} \pi (\cos x)^4 \, dx - \int_0^{\pi} x (\cos x)^4 \, dx$$

$$I = \int_0^{\pi} \pi \cos^4 x dx - I$$

$$[I = \int_0^{\pi} x \cos^4 x dx]$$

$$\text{Ans, } 2I = \int_0^{\pi} \pi \cos^4 x dx$$

28

$$\text{Ans, } 2I = 2 \int_0^{\pi/2} \pi \cos^4 x dx$$

by using

$$\text{Ans, } I = \pi \int_0^{\pi/2} \cos^4 x dx$$

$$= \pi \times \frac{3\pi}{16}$$

$$= \frac{3\pi^2}{16}$$

$$\left[\int_0^{\pi/2} \cos^4 x dx = \frac{3\pi}{16} \right]$$

$$x^b (x-a)^p \cos x \Big|_0^{\pi} - x^b (x-a)^p \cos x \Big|_0^{\pi} =$$

$$x^b \{ (x-a)^p \cos x \} \Big|_0^{\pi} - x^b \{ (x-a)^p \cos x \} \Big|_0^{\pi} =$$

$$x^b \{ (x-a)^p \cos x \} \Big|_0^{\pi} - x^b \{ (x-a)^p \cos x \} \Big|_0^{\pi} =$$

Example: 150. Find the circumference of a circle $x^2 + y^2 = R^2$ of radius R .

The equation of the circle is

$$x^2 + y^2 = R^2$$

$$\Rightarrow y^2 = R^2 - x^2$$

$$\Rightarrow y = \pm \sqrt{R^2 - x^2}$$

which means that,

as a graph, the upper semi-circle is $y = +\sqrt{R^2 - x^2}$

the lower " " " $y = -\sqrt{R^2 - x^2}$

We will find the length along this upper semi-circle, which is half the circumference of the circle. We have,

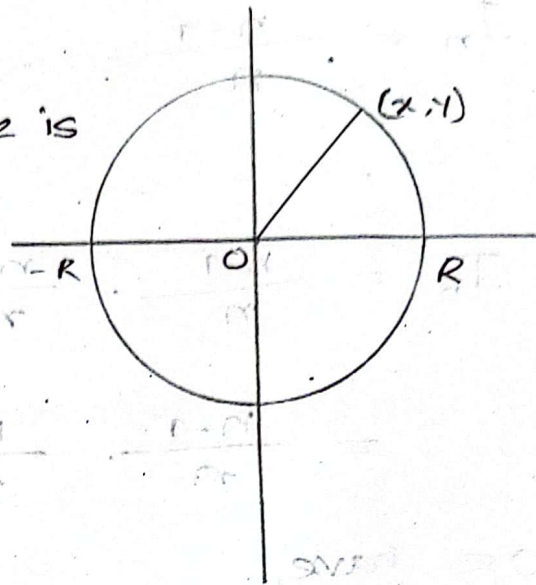
$$S = \int ds = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \quad \text{--- (1)}$$

Given,

$$y = f(x) = \sqrt{R^2 - x^2}$$

$$\Rightarrow \frac{dy}{dx} = f'(x) = (R^2 - x^2)^{-1/2}$$

$$\begin{aligned} \Rightarrow \frac{dy}{dx} = f'(x) &= \frac{1}{2} (R^2 - x^2)^{\frac{1}{2} - 1} \cdot \frac{d}{dx} (R^2 - x^2) \\ &= \frac{1}{2} (R^2 - x^2)^{-1/2} \cdot (0 - 2x) \end{aligned}$$



$$= (R^2 - x^2)^{-\frac{1}{2}} \cdot (-x)$$

$$\Rightarrow \frac{dy}{dx} = f'(x) = \frac{-x}{(R^2 - x^2)^{1/2}} \quad \text{--- (ii)}$$

Now,

$$\left(\frac{dy}{dx}\right)^2 = \{f'(x)\}^2 = \frac{(-x)^2}{\{\sqrt{R^2 - x^2}\}^2}$$

$$= \frac{x^2}{\{(R^2 - x^2)^{1/2}\}^2}$$

$$\therefore \left(\frac{dy}{dx}\right)^2 = \{f'(x)\}^2 = \frac{x^2}{(R^2 - x^2)} \quad \text{--- (iii)}$$

$$\Rightarrow 1 + \left(\frac{dy}{dx}\right)^2 = 1 + \{f'(x)\}^2$$

$$= 1 + \frac{x^2}{(R^2 - x^2)}$$

$$= \frac{R^2 - x^2 + x^2}{R^2 - x^2}$$

$$\therefore 1 + \left(\frac{dy}{dx}\right)^2 = 1 + \{f'(x)\}^2 = \frac{R^2}{(R^2 - x^2)} \quad \text{--- (iv)}$$

Then we calculate the arc length along the upper

Semi circle as from (i) $\rightarrow$

$$S = \int ds = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$= \int_{-R}^R \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$= \int_{-R}^R \sqrt{\frac{R^2}{R^2 - x^2}} dx$$

$$= \int_{-R}^R \frac{R}{\sqrt{R^2 - x^2}} dx$$

$$= R \int_{-R}^R \frac{1}{\sqrt{R^2 - x^2}} dx$$

$$= R \sin^{-1} \left[\frac{x}{R} \right]_{-R}^R$$

$$= R \sin^{-1} \left[\frac{R}{R} \right] - R \sin^{-1} \left[\frac{-R}{R} \right]$$

$$= R \sin^{-1} 1 - R \sin^{-1} (-1)$$

$$= R \sin^{-1} 1 + R \sin^{-1} 1$$

$$= R \sin^{-1} \sin \frac{\pi}{2} + R \sin^{-1} (\sin \frac{\pi}{2})$$

$$= R \frac{\pi}{2} + R \frac{\pi}{2} \therefore S = 2R \frac{\pi}{2} = R\pi$$

Hence the length of half circle is $R\pi$. So, the length of full circle is $2R\pi$. This means that the circumference of the circle is $C = 2\pi R$.

Example : 152. Find length of the arc AB of the curve with equation $y = \frac{2}{3}x^{\frac{3}{2}}$, where the x-co-ordinates of A and B are 3 and 8 respectively.

We have,

$$S = \int ds = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \quad \text{--- (1)}$$

Given,

$$y = f(x) = \frac{2}{3}x^{\frac{3}{2}}$$

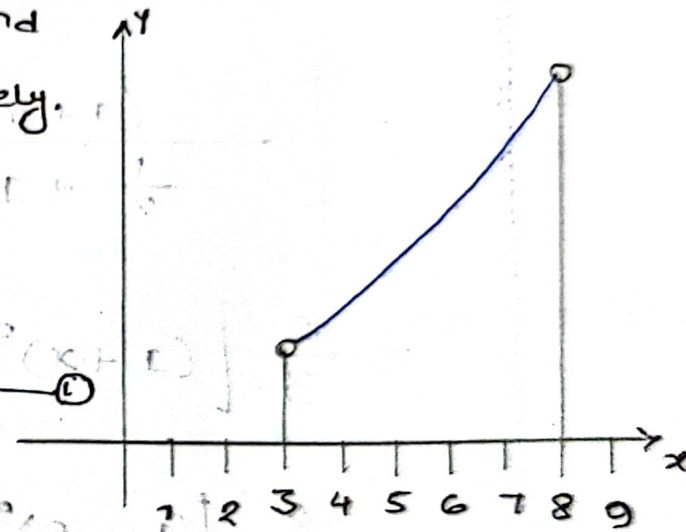
$$\Rightarrow \frac{dy}{dx} = \frac{2}{3} \times \frac{3}{2} x^{\frac{3}{2}-1}$$

$$= x^{\frac{3-2}{2}}$$

$$= x^{1/2}$$

$$\text{So, } 1 + \left(\frac{dy}{dx}\right)^2 = 1 + (x^{1/2})^2$$

$$= 1 + x$$



The length of the curve ^{AB} is: $\int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

Arce length AB $= \int_3^8 \left[1 + \left(\frac{dy}{dx}\right)^2 \right]^{1/2} dx$ [from ①]

$$= \int_3^8 (1+x)^{1/2} dx \quad \left[\because 1 + \left(\frac{dy}{dx}\right)^2 = 1 + (x^{1/2})^2 = 1+x \right]$$

$$= \left[\frac{(1+x)^{\frac{1}{2}+1}}{\frac{1}{2}+1} \right]_3^8$$

$$= \frac{2}{3} \left[(1+x)^{3/2} \right]_3^8$$

$$= \frac{2}{3} \left[(1+8)^{3/2} \right] - \frac{2}{3} \left[(1+3)^{3/2} \right]$$

$$= \frac{2}{3} \left[9^{3/2} \right] - \frac{2}{3} \left[4^{3/2} \right]$$

$$= \frac{2}{3} \left[(3^2)^{3/2} \right] - \frac{2}{3} \left[(2^2)^{3/2} \right]$$

$$= \frac{2}{3} (27 - 8)$$

$$= \frac{38}{3}$$

~~160~~ Example: 160. Let's try another example of an area bounded by a curve $y = x^3 + 1$, the x -axis and the limit of $x=0$ and $x=3$.

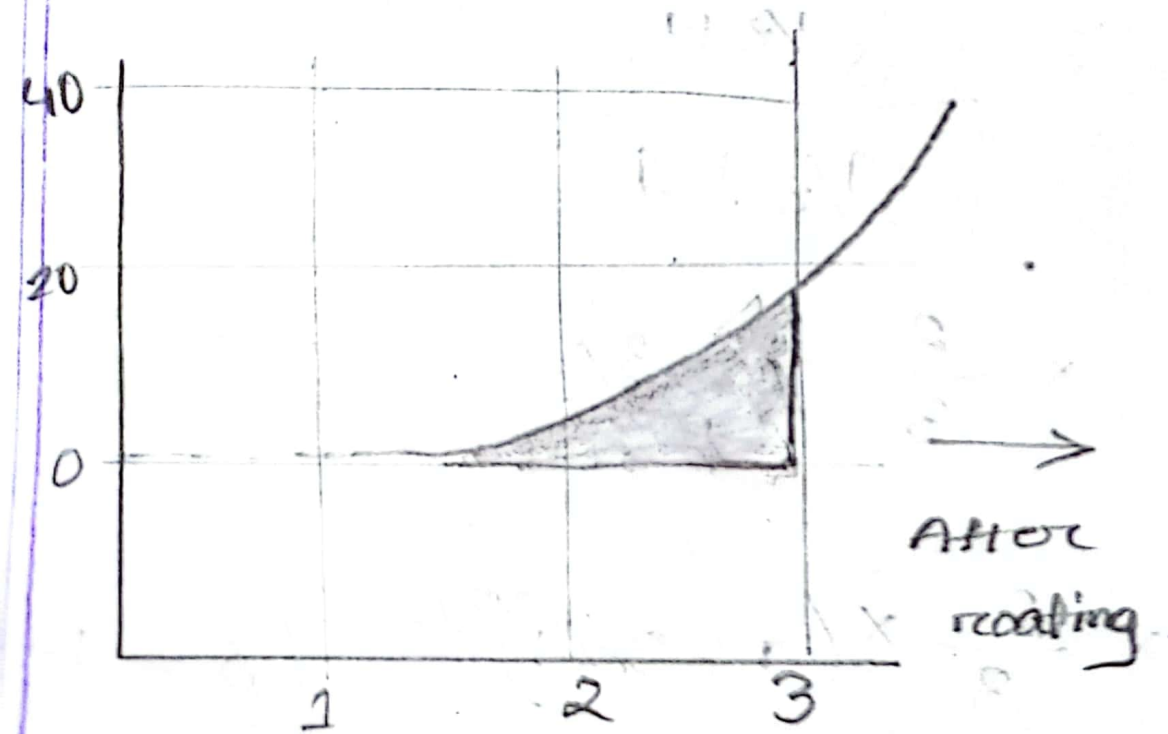


Fig-01

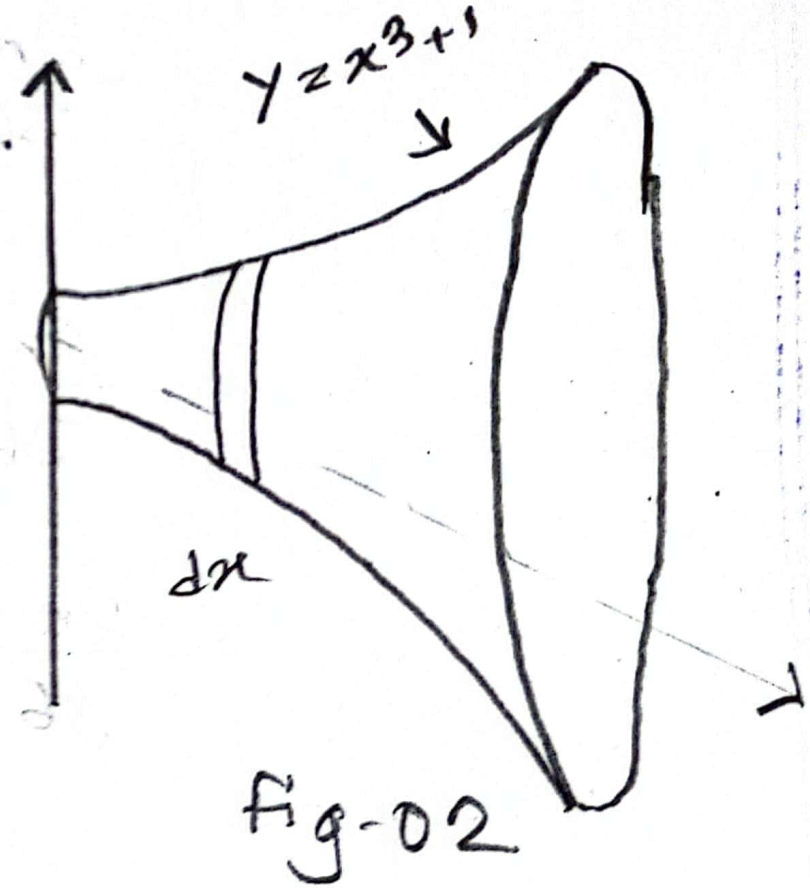


Fig-02

When the shaded area (Fig-01) is rotated 360° about the x -axis, we again observed that a volume (Fig-02) is generated.

Applying the above formula, we get,

$$\text{Volume} = \int_a^b \pi r^2 dx = \int_a^b \pi y^2 dx = \int_a^b \pi (f(x))^2 dx$$

Given,

$$y = x^3 + 1$$

$$= \pi \int_a^b y^2 dx$$

$$= \pi \int_0^3 (x^3 + 1)^2 dx$$

$$= \pi \int_0^3 (x^6 + 2x^3 + 1) dx$$

$$= \pi \left[\frac{x^7}{7} + \frac{2x^4}{2} + x \right]_0^3$$

$$= \pi ([35593] - 0)$$

$$= 1118.2 \text{ units}^3$$

$$y = \sqrt{x}$$

$$\text{Volume} = \int_1^4 \pi (y^2)^2 dx$$

Example: 162. Consider the area bounded by the straight line $y = 3x$, the x -axis and $x = 1$.

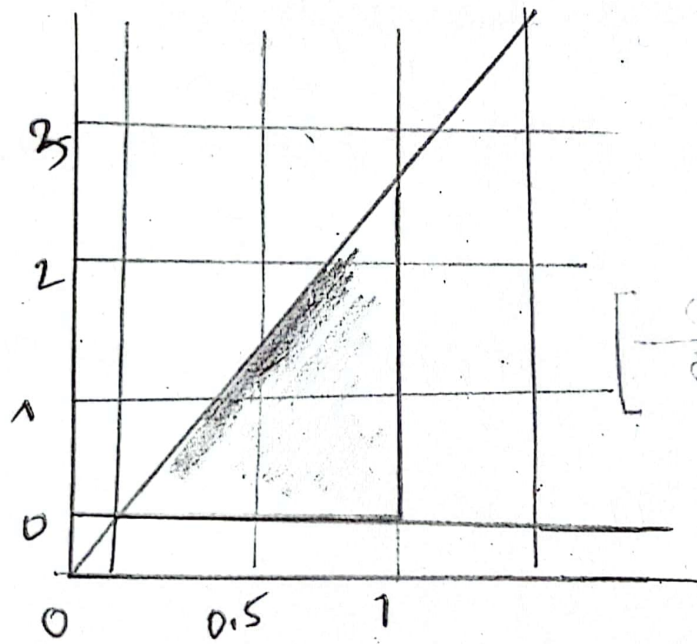


Fig-01

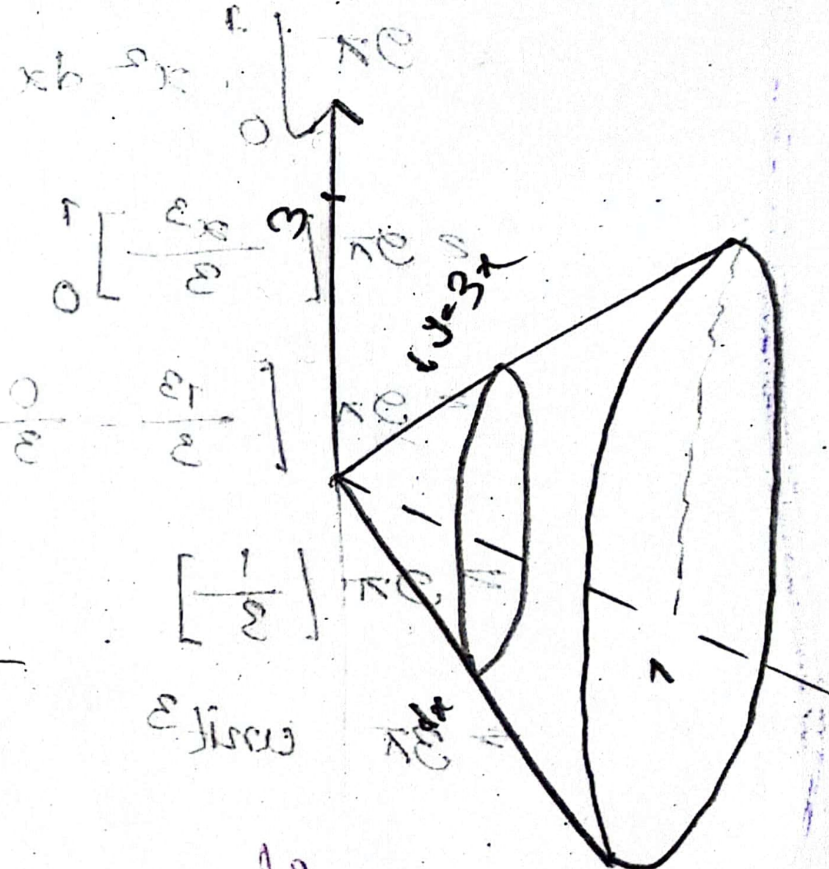


Fig-02 *

The resulting solid is a cone:

To find this volume we could take slices, each dx wide and radius y

Applying the above formula we get,

$$\text{Volume} = \int_a^b \pi r^2 dx = \int_a^b \pi y^2 dx = \int_a^b \pi (f(x))^2 dx$$

Given,

$$y = 3x$$

$$\text{Volume} = \int_0^1 \pi (3x^2)^2 dx$$

$$= \int_0^1 \pi 9x^4 dx$$

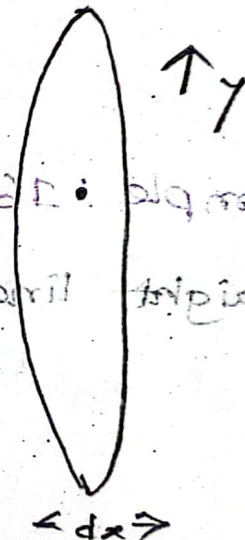
$$= 9\pi \int_0^1 x^4 dx$$

$$= 9\pi \left[\frac{x^5}{5} \right]_0^1$$

$$= 9\pi \left[\frac{1^5}{5} - \frac{0}{5} \right]$$

$$= 9\pi \left[\frac{1}{5} \right]$$

$$= 9\pi \text{ unit}^3$$



$$* V = \frac{\pi r^2 h}{3}$$

Example: 174. find the volume of the solid of revolution generated by rotating the curve $y = x^3$ between $y=0$ and $y=4$, about the y -axis.

$$\text{Volume} = \int_c^d \pi r^2 dy$$

$$= \int_c^d \pi x^2 dy$$

$$= \int_c^d \pi f(y)^2 dy$$

Given,

$$y = x^3$$

$$\Rightarrow y^{1/3} = (x^3)^{1/3}$$

$$\Rightarrow x = y^{1/3} = f(y)$$

And here is the volume generated,

from ① we get,

$$V = \int_c^d \pi x^2 dy = \int_c^d \pi x^2 dy$$

$$= \int_c^d \pi f(y)^2 dy$$

$$\text{Volume} = \int_0^4 \pi x^2 dy$$

$$= \int_0^4 \pi (y^{1/3})^2 dy$$

$$= \pi \int_0^4 y^{2/3} dy$$

$$= \pi \left[\frac{y^{2/3+1}}{2/3+1} \right]_0^4$$

$$= \pi \times \frac{3}{5} \left[y^{5/3} \right]_0^4$$

$$= \frac{3}{5} \pi \left[4^{5/3} - 0^{5/3} \right] = \frac{3}{5} \pi \times 4^{5/3}$$

CT Math

* Find the volume of the solid, generated by the revolution of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, with $(a > b)$, round it's minor axis, (the y -axis).

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\therefore \frac{x^2}{a^2} = 1 - \frac{y^2}{b^2}$$

$$\therefore \frac{x}{a} = \sqrt{1 - \frac{y^2}{b^2}}$$

$$\therefore x = a \sqrt{1 - \frac{y^2}{b^2}}$$

The volume of a disk at height y is,

$$dV = \pi \left(a \sqrt{1 - \frac{y^2}{b^2}} \right)^2 dy$$

$$= \pi a^2 \left(1 - \frac{y^2}{b^2} \right) dy$$

$$\therefore \text{Volume, } V = \int_{-b}^b \pi a^2 \left(1 - \frac{y^2}{b^2} \right) dy$$

$$= \pi a^2 \left[\int_{-b}^b 1 dy - \frac{1}{b^2} \int_{-b}^b y^2 dy \right]$$

$$= \pi a^2 \left[(y|_b^b) - \frac{1}{b^2} \left[\frac{y^3}{3} \right]_b^b \right]$$

$$= \pi a^2 \left[\left[b - (-b) \right] - \frac{1}{b^2} \left[\frac{b^3}{3} - \left(-\frac{b^3}{3} \right) \right] \right]$$

$$= \pi a^2 \left(2b - \frac{1}{b^2} \cdot \frac{2b^3}{3} \right)$$

$$= \pi a^2 \left(2b - \frac{2b}{3} \right)$$

$$= \pi a^2 \left(\frac{6b}{3} - \frac{2b}{3} \right)$$

$$= \pi a^2 \cdot \frac{4b}{3}$$

$$= \frac{4\pi a^2 b}{3}$$

$$V_b \left(\frac{s_b}{s_d} - 1 \right) \cdot s_{dN}$$

$$V_b \left(\frac{s_b}{s_d} - 1 \right) \cdot s_{dN}$$

Maisha_1AF

Example 163: Consider the solid of revolution formed by the graph of $y = x^2$ from $x = 0$ to $x = 2$:

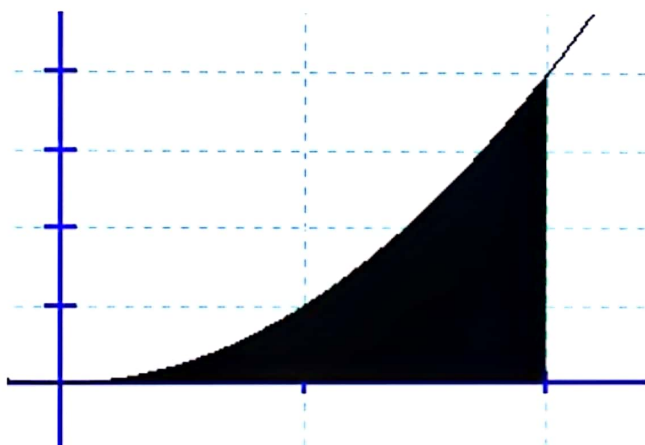


Figure # 59

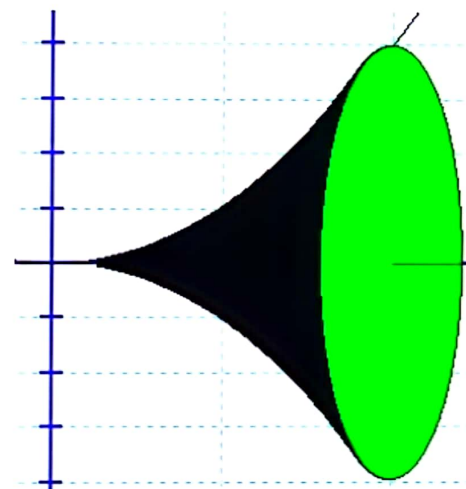
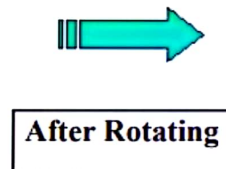


Figure # 60

Applying the above formula, we get,

$$\text{Volume} = \int_a^b \pi r^2 dx = \int_a^b \pi y^2 dx = \int_a^b \pi (f(x))^2 dx$$

Given, $y = x^2$

$$\text{Volume} = \int_0^2 \pi (x^2)^2 dx$$

$$\text{Volume} = \int_0^2 \pi x^4 dx = \pi \int_0^2 x^4 dx = \pi \left[\frac{x^5}{5} \right]_0^2$$

$$\text{Volume} = \frac{\pi}{5} [2^5 - 0] = \frac{\pi \times 2^5}{5} \text{ Answer}$$

6. If $y = \sin x$ in the range $0 \leq x \leq \pi$ is

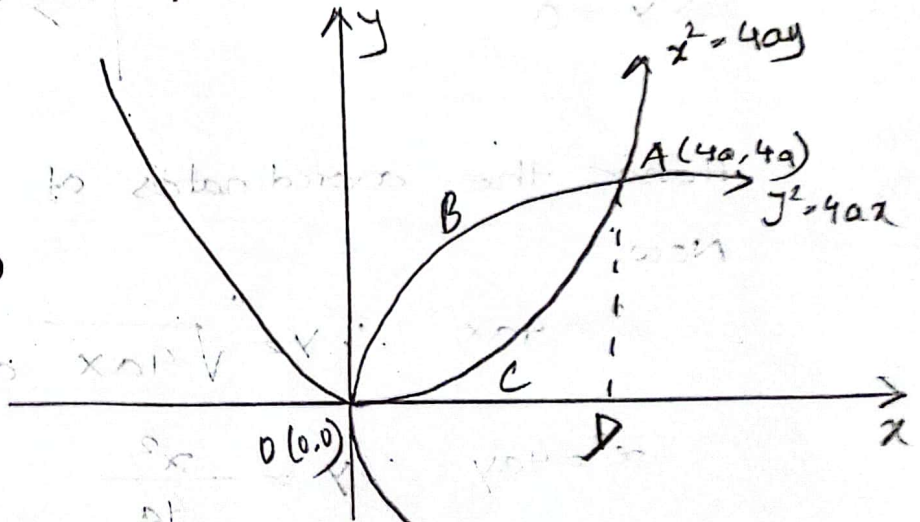
Quadrature

Example : 180. Find the area common to the two parabola $x^2 = 4ay$ and $y^2 = 4ax$

Given, Curve are

$$x^2 = 4ay \text{ ————— (I)}$$

$$y^2 = 4ax \text{ ————— (II)}$$



Since eq (I) contain only even power of x , so this curve is symmetrical about y -axis. and also curve (II) is symmetrical about x -axis.

From (I) and (II) have,

$$x^2 = 4ay$$

$$\Rightarrow (x^2)^2 = (4ay)^2$$

$$\Rightarrow x^4 = 16a^2y^2$$

$$\Rightarrow x^4 = 16a^2 \cdot 4ax \quad [y^2 = 4ax]$$

$$\Rightarrow x^4 - 16 \cdot 4a^3x = 0$$

$$\Rightarrow x(x^3 - 64a^3) = 0$$

$$\Rightarrow x = 0; \quad x^3 = 64a^3; \quad \text{i.e., } x = 4a.$$

Putting the value of x in (ii)

$$y^2 = 4ax$$

$$\Rightarrow y^2 = 4a \cdot 0 \quad [x=0]$$

$$\Rightarrow y = 0$$

$$y^2 = 4ax$$

$$\Rightarrow y^2 = 4a \cdot 4a$$

$$\Rightarrow y = 4a$$

Hence the coordinates of $O(0,0)$ and $A(4a,4a)$

Now,

$$y^2 = 4ax \quad \therefore y = \sqrt{4ax} \text{ and}$$

$$\therefore x^2 = 4ay \quad \therefore y = \frac{x^2}{4a}$$

Let,

$$y_1 = f_1(x) = \sqrt{4ax} \quad \text{--- (iii)}$$

$$y_2 = f_2(x) = \frac{x^2}{4a} \quad \text{--- (iv)}$$

We are to find the area of $OBAO$. It can be written as,

Area of $OBAO$ = area of $OBADO$ - area of $OADO$

[Upper curve - lower curve]

$$= \int_0^{4a} f_1(x) dx - \int_0^{4a} f_2(x) dx$$

$$= \int_0^{4a} y_1 dx - \int_0^{4a} y_2 dx$$

$$= \int_0^{4a} \sqrt{4ax} \, dx - \int_0^{4a} \frac{x^2}{4a} \, dx \quad [\text{iii} \& \text{iv}]$$

$$= 2\sqrt{a} \left[\frac{2}{3} x^{3/2} \right]_0^{4a} - \frac{1}{4a} \left[\frac{x^3}{3} \right]_0^{4a}$$

$$= \frac{4\sqrt{a}}{3} (4a)^{3/2} - 0 - \frac{1}{4a} \left[\frac{(4a)^3}{3} - 0 \right]$$

$$= \frac{4\sqrt{a}}{3} (2^2)^{3/2} \cdot a^{3/2} - \frac{1}{4a} \cdot \frac{(2^2)^3 \cdot a^3}{3} = \frac{4\sqrt{a}}{3} \cdot 2^3 \cdot a^{3/2} - \frac{16a^2}{3}$$

$$= \frac{4\sqrt{a}}{3} 2^3 \cdot a^{3/2} - \frac{16a^2}{3}$$

$$= \frac{4a^{1/2}}{3} \cdot 8 \cdot a^{3/2} - \frac{16a^2}{3}$$

$$= \frac{32}{3} a^{1/2 + 3/2} - \frac{16a^2}{3}$$

$$= \frac{32}{3} a^2 - \frac{16}{3} a^2$$

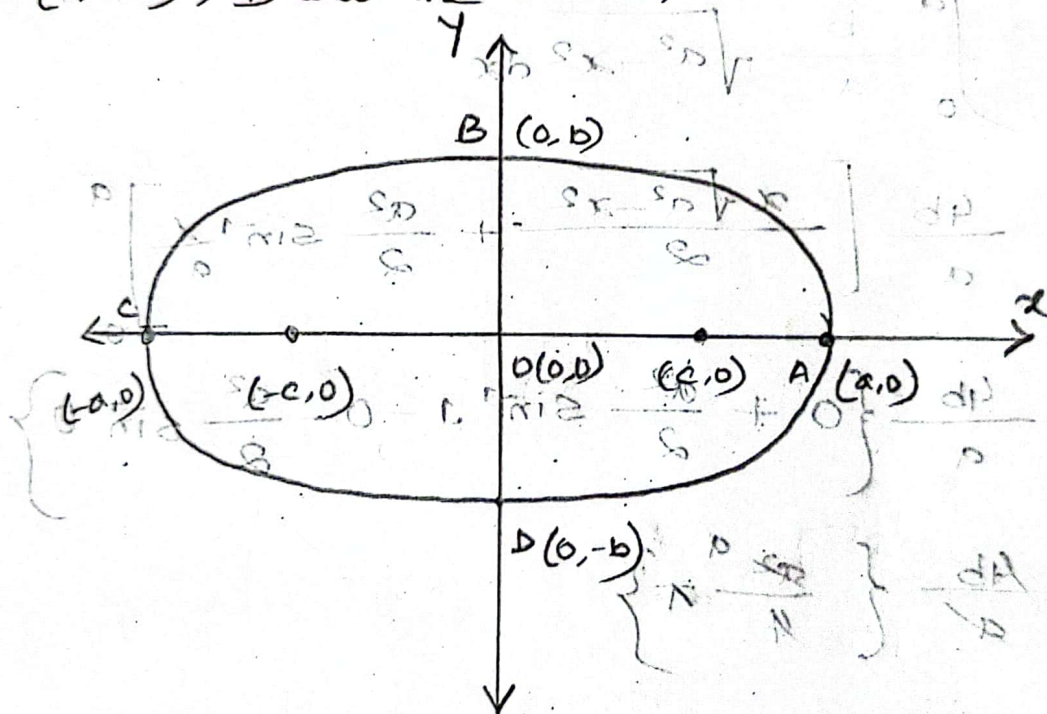
$$= \frac{16}{3} a^2 \text{ square units.}$$

Example : 186. Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Given,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{--- (1)}$$

The curve is symmetrical about the both axis and it meets the axis at points $A(a, 0)$, $B(0, b)$, $C(-a, 0)$, and $D(0, -b)$. Draw the curve,



From (1) we have,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\Rightarrow \frac{y^2}{b^2} = 1 - \frac{x^2}{a^2}$$

$$\Rightarrow \frac{y^2}{b^2} = \frac{a^2 - x^2}{a^2}$$

$$\Rightarrow y^2 = \frac{a^2 - x^2}{a^2} b^2$$

$$\Rightarrow y = \pm \frac{b}{a} \sqrt{a^2 - x^2}$$

Now the area of the ellipse is = 4 × (area of ABDA)

$$= 4 \int_0^a y \, dx$$

$$= 4 \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} \, dx$$

$$= \frac{4b}{a} \left[\frac{x\sqrt{a^2 - x^2}}{2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} \right]_0^a$$

$$= \frac{4b}{a} \left\{ 0 + \frac{a^2}{2} \sin^{-1} 1 - 0 - \frac{a^2}{2} \sin^{-1} 0 \right\}$$

$$= \frac{4b}{a} \left\{ \frac{a^2}{4} \pi \right\}$$

$$= ab\pi$$

Therefore the area of the ellipse is $ab\pi$.

square unit.

Triple Integration

Math

#83

$$\begin{aligned} & \int_0^1 \int_0^2 x^2 dx dy \\ &= \int_0^1 \left[\frac{x^3}{3} \right]_0^2 dy \\ &= \int_0^1 \left[\frac{2^3}{3} \right] dy \\ &= \int_0^1 \frac{8}{3} dy \\ &= \frac{8}{3} \int_0^1 dy \\ &= \frac{8}{3} (1-0) \\ &= \frac{8}{3} \end{aligned}$$

$$\begin{aligned} & \# \int_0^1 \int_0^3 \int_0^2 2x dz dy dx \\ &= \int_0^1 \int_0^3 \left[\int_0^2 2x dz \right] dy dx \\ &= \int_0^1 \int_0^3 [2x(2-0)] dy dx \\ &= \int_0^1 \int_0^3 4x dy dx \\ &= \int_0^1 4x(3-0) dx \\ &= \int_0^1 12x dx \end{aligned}$$

$$\begin{aligned} \text{OR, } & \# \int_0^2 \int_0^1 x^2 dy dx \\ &= \int_0^2 [x^2(1-0)] dx \\ &= \int_0^2 x^2 dx \\ &= \left[\frac{x^3}{3} \right]_0^2 \\ &= \frac{2^3}{3} \\ &= \frac{8}{3} \end{aligned}$$

$$\begin{aligned} &= 12 \int_0^1 x dx \\ &= 12 \left[\frac{x^2}{2} \right]_0^1 \\ &= 12 \cdot \frac{1}{2} \\ &= 6 \end{aligned}$$

Triple integral

Evaluate

$$\begin{aligned}
 & \int_0^1 \int_0^{1-x} \int_0^{1-x-y} z \, dz \, dy \, dx \\
 &= \int_0^1 \int_0^{1-x} \left[\frac{z^2}{2} \right]_0^{1-x-y} dy \, dx \\
 &= \int_0^1 \int_0^{1-x} \left[\frac{(1-x-y)^2}{2} \right] dy \, dx \quad \left[\int x^n dx = \frac{x^{n+1}}{n+1} \right] \\
 &= \int_0^1 \int_0^{1-x} \left[\frac{(1-x)^2 - 2(1-x)y + y^2}{2} \right] dy \, dx \\
 &= \int_0^1 \left[\frac{(1-x)^2 y - (1-x)y^2 + \frac{y^3}{3}}{2} \right]_0^{1-x} dx \\
 &= \int_0^1 \left[\frac{(1-x)^2 (1-x) - (1-x)(1-x)^2 + \frac{(1-x)^3}{3}}{2} \right] dx \\
 &= \frac{1}{2} \int_0^1 \left[(1-x)^3 - (1-x)^3 + \frac{(1-x)^3}{3} \right] dx \\
 &= \frac{1}{2} \int_0^1 \left[\frac{(1-x)^3}{3} \right] dx \\
 &= \frac{1}{2} \left[\frac{(1-x)^4}{4} \right]_0^1 \\
 &= \frac{1}{2} \left[\frac{(1-1)^4}{4} - \frac{(1-0)^4}{4} \right] \\
 &= \frac{1}{2} \left[0 - \frac{1}{4} \right] \\
 &= -\frac{1}{8}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2} \int_0^1 \left[(1-x) - 0 \right] - 2 \left[\frac{(1-x)^3}{3} - \frac{0^3}{3} \right] + \left[\frac{(1-x)^5}{5} - \frac{0^5}{5} \right] dx \\
 &= \frac{1}{2} \int_0^1 \left[(1-x) - 2 \left[\frac{(1-x)^3}{3} \right] + \left[\frac{(1-x)^5}{5} \right] \right] dx \\
 &= \frac{1}{2} \left[\int_0^1 (1-x) dx - \frac{2}{3} \int_0^1 (1-x)^3 dx + \frac{1}{5} \int_0^1 (1-x)^5 dx \right] \\
 &= \frac{1}{2} \left[\int_0^1 1 dx - \int_0^1 x dx - \frac{2}{3} \int_0^1 (1-x)^3 dx + \frac{1}{5} \int_0^1 (1-x)^5 dx \right] \\
 &= \frac{1}{2} \left[\left[x \right]_0^1 - \left[\frac{x^2}{2} \right]_0^1 - \frac{2}{3} \left[\frac{(1-x)^{3+1}}{3+1} (-1) \right]_0^1 + \frac{1}{5} \left[\frac{(1-x)^{5+1}}{5+1} (-1) \right]_0^1 \right] \quad \left[\int dx = x \right] \\
 &= \frac{1}{2} \left[\left[1 - 0 \right] - \left[\frac{1^2}{2} - \frac{0^2}{2} \right] + \frac{2}{3} \left[\frac{(1-x)^4}{4} \right]_0^1 - \frac{1}{5} \left[\frac{(1-x)^6}{6} \right]_0^1 \right] \\
 &= \frac{1}{2} \left[\left[1 - 0 \right] - \left[\frac{1^2}{2} - \frac{0^2}{2} \right] + \frac{2}{3} \left[\frac{(1-1)^4}{4} - \frac{(1-0)^4}{4} \right] - \frac{1}{5} \left[\frac{(1-1)^6}{6} - \frac{(1-0)^6}{6} \right] \right] \\
 &= \frac{1}{2} \left[\left[1 \right] - \left[\frac{1}{2} \right] + \frac{2}{3} \left[0 - \frac{1}{4} \right] - \frac{1}{5} \left[0 - \frac{1}{6} \right] \right] \\
 &= \frac{1}{2} \left[1 - \frac{1}{2} + \frac{2}{3} \left[-\frac{1}{4} \right] - \frac{1}{5} \left[-\frac{1}{6} \right] \right] \\
 &= \frac{1}{2} \left[1 - \frac{1}{2} - \frac{2}{12} + \frac{1}{30} \right] \\
 &= \frac{1}{2} \left[\frac{60 - 30 - 10 + 2}{60} \right] \\
 &= \frac{1}{2} \left[\frac{22}{60} \right] \\
 &= \frac{11}{60}
 \end{aligned}$$